

Supplementary Information

A flexible modelling framework for estimating thermal sensitivity across life

Daniel W.A. Noble^{*1}, Pieter A. Arnold¹, Patrice Pottier²

¹*Division of Ecology and Evolution, Research School of Biology, Australian National University, Canberra, Australia*

²*Department of Biological and Environmental Sciences, University of Gothenburg, Gothenburg, Sweden*

today

1. The thermal load sensitivity (TLS) framework is the gold standard for estimating and predicting thermal tolerance in nature, but its standard two-step pipeline — point-estimate LT50 extraction at each assay temperature followed by ordinary least squares regression of $\log_{10}$ LT50 on temperature — rests on statistical assumptions that are routinely violated and that bias every downstream quantity (z , CTmax, heat-injury predictions).
2. We describe a flexible Bayesian, single-stage four-parameter logistic (4PL) model for modelling the impact of temperature and time on survival and fertility. We demonstrate how such an approach can recover all the classical TDT output typical of a two-stage modelling framework — including thermal sensitivity (z , slopes), critical thermal limit ($CT_{max_{1hr}}$) and the heat-injury (HI) integral, which allows for one to predict thermal injury in nature under realistic exposure temperatures. Given its fully Bayesian implementation, it allows for error propagation to account for uncertainty at all stages of modelling as well as improved inferences.
3. We applied our new modelling framework to data from brown shrimp (*Crangon crangon*) and XX to demonstrate the equivalencies and benefits of such an approach.
4. We show that the classical and Bayesian frameworks are equivalent, but that the hierarchical 4PL delivers properly propagated uncertainty, can account for complex hierarchical data (i.e., random effects), can accommodate censoring and overdispersion, , and unlocks outputs the two-step pipeline cannot provide (full survival surface, arbitrary mortality percentiles, decomposable between-group contrasts, dynamic-regime predictions). We provide reusable R helpers and recommend the hierarchical 4PL as the default analysis for new TDT data.

^{*}Corresponding author: daniel.noble@anu.edu.au

Table of contents

1	Introduction	2
2	A non-linear Bayesian hierarchical model for characterizing thermal load sensitivity	3
2.1	Model specification	3
3	Deriving key thermal load sensitivity parameters	4
4	Predicting heat injury and survival under dynamic thermal environments	5
5	Application of Bayesian dose-response models to the TLS framework	6
6	Case Study 1: Thermal sensitivity estimation across life stages in shrimp	8
6.1	What the 4PL delivers that the two-step cannot	8
7	Worked example: brown shrimp thermal tolerance	8
7.1	Data and experimental design	8
7.2	Lethal TDT — hierarchical 4PL on counts	8
7.3	Sublethal time-to-death	8
7.4	Dynamic CTmax across ramping rates	8
7.5	Reconciling the three endpoints via the Jørgensen ramping equation	8
7.6	Field-temperature heat-injury projections under warming scenarios	8
8	Discussion	8
8.1	What changes for the field if the hierarchical 4PL becomes the default	8
8.2	Limits of the present analysis	8
8.3	Recommendations for primary studies and meta-analyses	8
9	Conclusions	8
10	Acknowledgements	8
	Author contributions	8
	Data and code availability	8
	Conflict of interest	9
	References	9
11	A Tutorial with Simulations	11
11.0.1	<i>Simulating Data</i>	11
11.0.2	<i>The classical two-stage TDT pipeline</i>	14
11.0.3	<i>A joint Bayesian 4PL</i>	16
11.0.4	<i>Deriving z and $CT_{max_{1hr}}$ from the posterior</i>	20
11.0.5	<i>Comparing the two pipelines</i>	24
11.0.6	<i>Extending the model: temperature-varying shape parameters</i>	24
11.0.7	<i>Heat injury accumulation and survival under a realistic field trace</i>	36

1 Introduction

Climate change is driving more frequent and extreme heat events globally, compromising organism fitness (Arnold *et al.* 2025; Rezende *et al.* 2020; Snook *et al.* 2026), with cascading consequences for population persistence and biodiversity (Smale *et al.* 2019; Wiens 2016). Quantifying differences in thermal tolerance across species is now central to forecasting climate-change impacts on organisms in nature (Deutsch *et al.* 2008; Jørgensen *et al.* 2022; Pinsky *et al.* 2019; Sunday *et al.* 2011, 2012). Static measures of heat tolerance, such as CT_{max} , have long been used to establish differences in vulnerability among taxa and predicting climate change risk (Deutsch *et al.* 2008; Pinsky *et al.* 2019; Pottier *et al.* 2025; Sunday *et al.* 2011, 2012). While useful, static measures do not capture how the negative effects of extreme heat on fitness depend on both heat intensity and duration, limiting our ability to make realistic predictions under more dynamic thermal environments (Jørgensen *et al.* 2021; Ørsted *et al.* 2022; Rezende *et al.* 2014).

The Thermal Load Sensitivity (TLS) framework (or equivalently Thermal Death Time framework) (Arnold *et al.* 2025; Jørgensen *et al.* 2019, 2022; Noble *et al.* 2026; Ørsted *et al.* 2022; Rezende *et al.* 2014) resolves these limitations by jointly modelling the effects of heat intensity and duration on fitness (e.g., survival, fertility, reproduction). This matters because physiological damage accumulates in a dose-dependent manner — physiological damage accumulates faster at higher temperatures and over longer exposures (Arnold *et al.* 2025; Jørgensen *et al.* 2019; Ørsted *et al.* 2022; Rezende *et al.* 2014). Organisms experiencing fluctuating thermal environments in nature can therefore accumulate damage to physiological systems that results in sublethal injury that compounds across successive heat exposures if not fully repaired (Arnold *et al.* 2025; Buckley *et al.* 2025; Jørgensen *et al.* 2021). Such processes can lead to delayed responses to extreme heat events.

Thermal load sensitivity modelling, while powerful, traditionally involves a two-step approach (Jørgensen *et al.* 2019; Li *et al.* 2023; Ørsted *et al.* 2022; Rezende *et al.* 2020; Truebano *et al.* 2018). First, experiments will typically be done that manipulate a range of temperatures and times (5 temperatures x 5 durations). The proportion of live/dead (or fertile/infertile, if sub-lethal) is then modelled as a function of log transformed time (in minutes or hours) using dose-response curves fit across all temperatures (usually at each temperature treatment independently). A threshold response, e.g., LT_{50} or LT_{80} , is then derived from these curves. Second, the log transformed time taken to reach the threshold (e.g., LT_{50}) and temperature are regressed against each other using a linear model and thermal sensitivity (i.e., the slope of this regression, z) and critical thermal maxima (i.e., the intercept of the regression, $CT_{max_{time}}$) are then derived (Ørsted *et al.* 2022; Rezende *et al.* 2014). $CT_{max_{time}}$ is usually computed at 1 min or 1 hour duration. The 1 hour reference falls within typical assay durations avoiding substantial extrapolation which is why it is recommended (Jørgensen *et al.* 2019). Thermal sensitivity and $CT_{max_{1min}}$ are then used to quantify Heat Injury (HI) (e.g., Jørgensen *et al.* (2019); Villeneuve & White (2024)) in response to realistic temperatures in nature. Heat injury measures the cumulative percentage or probability until reaching a given threshold (e.g., LT_{50}), at which point we would predict 50% of the population to have died. Recent

work has emphasized the importance of repair in heat injury dynamics (Arnold *et al.* 2025; Buckley *et al.* 2025) with modelling approaches developed to capture these damage-repair dynamics.

While the two-step approach to modelling thermal sensitivity is intuitive and simple in practice it has numerous practical and statistical limitations. First, the two-step pipeline propagates uncertainty incorrectly. The LT_{50} estimates are estimates with non-negligible sampling variance. Treating them as fixed inputs to the regression discards that variance which can lead to miscalibrated standard errors on the slope (z) and intercept (CT_{max}) – a known issue in the statistical literature (Burke *et al.* 2017; Carroll *et al.* 2006). Second, group comparisons (between species, sexes, or life stages) inherit the miscalibrated uncertainty of the first stage. As such, differences in z between groups cannot be attributed to differences in the upper asymptote, lower asymptote, or curve steepness of the dose-response curve, because all three are collapsed into a single LT_{50} before the comparison is made. Third, the LT_{50} -extraction step collapses the within-replicate count, foreclosing hierarchical decomposition of variance across replicates, plates, populations, sexes, and life stages that could affect parameters of dose-response curves (Harrison 2014; Harrison *et al.* 2018; Nakagawa & Santos 2012). Fourth, the intrinsically hierarchical structure of (e.g., animals within batches and assay days) is generally ignored which can lead to pseudoreplication, resulting in anti-conservative standard errors that affect inference. Finally, propagating uncertainty from a two-step fit into downstream predictions — such as cumulative heat injury — is rarely attempted in practice understating the credible range of extreme heat impacts in nature.

Here, we present an alternative to the two-step pipeline: a single hierarchical four-parameter logistic (4PL) model fit jointly to all temperature by duration data. The 4PL is a flexible sigmoidal form that pairs with different sampling distributions — binomial for counts of dead or infertile individuals, time-to-event for latencies (e.g., time to knockdown), Beta for continuous proportions, or Gaussian for continuous performance measures (Nielsen *et al.* 2004; Ritz 2010; Ritz *et al.* 2015; Ritz & Streibig 2008; Rudemo *et al.* 1989). We focus on binomial outcomes for clarity, since these dominate the TLS literature (Ørsted *et al.* 2022, 2024; Rezende *et al.* 2020), but the derivations below apply unchanged when the likelihood is swapped. Additionally, while we use the four-parameter log-logistic form throughout for its flexibility other monotonic sigmoidal families (e.g., Weibull, log-normal) and simpler nested versions of the log-logistic (3PL, with one asymptote fixed; 2PL, with both fixed at 0 and 1) can be used within this framework (Ritz *et al.* 2015; Ritz & Streibig 2008). We show how all the classical thermal load sensitivity parameters (z , CT_{max}) can be derived from this single model, while also easily allowing for predictions of heat injury accumulation and survival. We develop these models within a Bayesian framework which allows for better uncertainty calibration and error propagation with complex hierarchical data structures.

2 A non-linear Bayesian hierarchical model for characterizing thermal load sensitivity

2.1 Model specification

Assuming we have mortality data for different times and temperatures, let y_{ij} denote the number of survivors out of n_{ij} individuals in trial j exposed for time t_{ij} (e.g., minutes) at assay temperature T_{ij} (°C). We can model the counts directly with a binomial likelihood (but a beta-binomial likelihood function can also be used and controls for overdispersion):

$$y_{ij} \mid n_{ij}, p_{ij} \sim \text{Binomial}(n_{ij}, p_{ij}), \quad (1)$$

with survival probability p_{ij} given by a four-parameter logistic (4PL) function of $\log_{10}$ -exposure time,

$$p_{ij} = \ell + \frac{u - \ell}{1 + \exp(k(\log_{10} t_{ij} - \text{mid}_{ij}))}. \quad (2)$$

where ℓ is the lower survival probability asymptote, u is the upper survival probability asymptote, and k is the slope defining the change in survival probability on the $\log_{10} t$ axis. Larger k gives a sharper decline in survival probability with $\log_{10} t$. mid_{ij} is the value of $\log_{10} t$ at which p falls to the midpoint between ℓ and u . When asymptotes are near 0 and 1, mid_{ij} coincides with $\log_{10} \text{LT50}$ at trial j . While other logistic forms can be used (e.g., two or 3 parameter logistics), Equation 2 is the most flexible model that can account for different asymptotic survival probabilities.

The key parameters of Equation 2 (i.e., $\ell, u, k, \text{mid}_{ij}$), can be fit as a non-linear function within a likelihood or Bayesian framework, which estimates their correlation and allows for fixed or random effects to be estimated to explain variation in these parameters. Such an approach allows for *a priori* comparisons (e.g., do sexes or species differ in midpoints?), with correct inference (e.g., do species differ significantly from each other?) and allows for variance decomposition to help identify key sources of variance in the data (e.g., estimating variances explained across individuals, species, trails etc.). Such models are most flexibly fit within a Bayesian framework with intuitive interpretation of these parameters and a highly flexible path for error propagation. We show below how such a model can be used to derive all thermal load sensitivity parameters of interest to ecophysiologists.

3 Deriving key thermal load sensitivity parameters

The TLS framework is elegant in that it derives metrics with clear biological interpretation that can be compared across populations and species (e.g., $z, CT_{\text{max}_{1hr}}$). Here, we show that we can derived these same metrics directly from Equation 2.

As indicated above, parameters can be fit to model changes in the mean of any of the four parameters in Equation 2. We can easily derive thermal sensitivity (z) and $CT_{\text{max}_{1hr}}$ by modelling the effect of temperature on midpoints as follows:

$$\text{mid}_{ij} = \beta_0 + \beta_1 (T_{ij} - \bar{T}), \quad (3)$$

where mid_{ij} is the midpoint ($\log_{10}$ -time at 50% survival between ℓ and u) for trial j at assay temperature T_{ij} ($^{\circ}\text{C}$); β_0 is the intercept giving the midpoint at the grand mean temperature $\bar{T}$; β_1 is the slope describing how the midpoint changes per $^{\circ}\text{C}$ increase in T_{ij} ; and $\bar{T}$ is the grand mean of assay temperatures used to mean-centre T_{ij} , which decorrelates β_0 from β_1 and improves parameter estimation and interpretation (Schielzeth 2010).

Setting $p_{ij} = 0.5$ in Equation 2, isolating the exponential, and taking natural logs gives $\log_{10} \text{LT50}(T) = \text{mid}(T) + (1/k) \log[(u - 0.5)/(0.5 - \ell)]$. Substituting into Equation 3 and grouping all terms that do not depend on T into an intercept, α , recovers the classical TDT linear relation:

$$\log_{10} \text{LT50}(T) = \underbrace{\left[\beta_0 + \frac{1}{k} \log \frac{u-0.5}{0.5-\ell} - \beta_1 \bar{T} \right]}_{\alpha} + \beta_1 T, \quad (4)$$

Note that Equation 4 subtracts $\beta_1 \bar{T}$ in α because we mean-centred T . With this subtraction, the intercept estimates $\log_{10} \text{LT50}$ at the average temperature ($\bar{T}$); without it, the intercept estimates $\log_{10} \text{LT50}$ at $T = 0^\circ\text{C}$. From Equation 4, the classical TLS quantities fall out directly. Thermal sensitivity is $z = -1/\beta_1$, and the critical thermal limit at any reference time t_{ref} is $CT_{\text{max}}(t_{\text{ref}}) = (\log_{10} t_{\text{ref}} - \alpha)/\beta_1$ (with $t_{\text{ref}} = 60$ min for $CT_{\text{max}_{hr}}$). Each of these quantities can be estimated separately for different strata (e.g., sex, population, species) by including the relevant fixed covariates and/or random effects in the linear predictor for mid_{ij} — and, if appropriate, in the linear predictors for ℓ , u , and k , each of which is modelled separately in the joint Bayesian fit (see Equation 12 below).

Importantly, the slope β_1 — and hence z — is robust to estimated asymptotes because the asymmetry correction $\frac{1}{k} \log \frac{u-0.5}{0.5-\ell}$ does not depend on T . Instead, it shifts $\log_{10} \text{LT50}$ vertically without changing the slope. In contrast, threshold summaries are not robust in this way; when partial mortality or sparse data do not estimate the asymptotes to be around 0 and 1, the asymmetry correction must be retained — added to the naive $\log_{10} \text{LT50}$ (Equation 5) and subtracted, after division by the slope (to convert the vertical shift in $\log_{10} \text{LT50}$ into the corresponding shift in temperature), from the naive CT_{max} (Equation 6):

$$\log_{10} \text{LT50}(T) = \underbrace{\beta_0 + \beta_1(T - \bar{T})}_{\text{naive estimate}} + \underbrace{\frac{1}{k} \log \frac{u-0.5}{0.5-\ell}}_{\text{correction}}, \quad (5)$$

$$CT_{\text{max}}(t_{\text{ref}}) = \underbrace{\frac{\log_{10} t_{\text{ref}} - \beta_0 + \beta_1 \bar{T}}{\beta_1}}_{\text{naive estimate}} - \underbrace{\frac{1}{\beta_1} \cdot \frac{1}{k} \log \frac{u-0.5}{0.5-\ell}}_{\text{correction}}. \quad (6)$$

In practice, however, these corrections may not be needed so long as the experimental design includes treatments that span the full mortality range which anchors ℓ and u tightly near 1 and 0 and makes the correction term effectively zero. When this is the case, $\log_{10} \text{LT50}$ and CT_{max} can be read directly from the naive forms in Equation 5 and Equation 6. Regardless, we recommend using the correction factor as a safeguard. However, computing both the naive and corrected estimates and comparing them is a useful sensitivity check for whether the correction is needed in any given dataset.

4 Predicting heat injury and survival under dynamic thermal environments

A particularly powerful aspect of TLS models are their ability to translate dynamic temperature times series' from nature to predict heat injury (HI) accumulation and thus population survival probability (Jørgensen *et al.* 2021; Li *et al.* 2023; Ørsted *et al.* 2022, 2024; Rezende *et al.* 2020). Heat injury accumulates as the time spent at a given temperature divided by the median lethal time at that temperature, integrated across the temperature trajectory $T(i)$ as follows:

$$HI(t) = 100 \int_0^t 10^{(T(i) - CT_{max_{1hr}})/z} di, \quad (7)$$

where z is the thermal sensitivity and $CT_{max_{1hr}}$ is the static temperature at which 50% of the population reaches the assayed endpoint after one hour of exposure. When $HI(t) = 100\%$, the cumulative dose equals one LT_{50} at the 1-hour reference, predicting 50% mortality (or 50% loss of the assayed sublethal endpoint, e.g. fertility; Ørsted *et al.* (2024)). Injury accumulates only when $T(i)$ exceeds a critical temperature T_c below which damage is repaired as fast as it accrues (Jørgensen *et al.* 2021; Ørsted *et al.* 2022). Field temperature records are typically available at hourly resolution. Equation 7 can then be evaluated by summing $100 \cdot 10^{(T_i - CT_{max_{1hr}})/z}$ across each hourly interval whose temperature T_i exceeds T_c , following the implementations in Jørgensen *et al.* (2021), Ørsted *et al.* (2024), and Li *et al.* (2023). To propagate uncertainty, z and $CT_{max_{1hr}}$, calculated from our Bayesian dose-response model can be used to calculate $HI(t)$ through time by substituting samples from the posterior distribution for each parameter into Equation 7.

A more natural prediction to make may instead be the population-level survival probability over time. This has the advantage of allowing for a direct connection to population dynamical models which typically use survival, growth and fecundity as vital rates for population-level predictions (Noble *et al.* 2026). To predict survival, we can calculate the *effective exposure time* at a chosen reference temperature T_{ref} . More effective time is accumulated at hot temperatures and less at cool ones, in proportion to the scaling factor $10^{(T(i) - T_{ref})/z}$,

$$\tau_{eff}(t) = \int_0^t 10^{(T(i) - T_{ref})/z} di, \quad (8)$$

We can then evaluate the 4PL surface once at the cumulative effective exposure to obtain the predicted population-level survival fraction at every point in the trajectory:

$$S_{pred}(t) = S(T_{ref}, \tau_{eff}(t)). \quad (9)$$

Here $S(T, \tau)$ is the survival surface fitted by the dose-response model, so $S_{pred}(t)$ is the survival probability for an organism held at *constant* T_{ref} after exposure duration $\tau_{eff}(t)$.

If calculating HI , it is important to realise that the Bayesian dose-response model can also be used directly to calculate HI at any survival threshold that is of interest (Jørgensen *et al.* 2021; Rezende *et al.* 2020). For example, the same posterior samples of $(\ell, u, k, m(T))$ give access to any percentile LT_x (the time at which $x\%$ of the population reaches the assayed endpoint) as follows:

$$\log_{10} LT_x(T) = m(T) - \frac{1}{k} \ln \frac{u - x/100}{x/100 - \ell}, \quad (10)$$

Substituting $LT_x(T)$ for $LT_{50}(T)$ in the HI equation returns HI_x for any survival threshold of interest. For example, relaxing x with 0.10 (i.e., LT_{10}) can be used to calculate the onset of early levels of mortality which maybe good for risk management or LT_{90} which might be more useful for understanding population-collapse thresholds. In each case, uncertainty in any predictions can always be presented alongside population-level averages or medians.

5 Application of Bayesian dose-response models to the TLS framework

We used a four-parameter logistic function to model the impact of temperature and duration (hours/minutes) on survival probability thermal sensitivity, estimate CT_{max1h}

To break the well-known likelihood ridges among (ℓ, u, k) in unconstrained 4PL fits (Bolker 2008; Gelman 2006), we estimate the identifiable parameterisation

$$\ell = 0.49 \logit^{-1}(\tilde{\ell}), \quad u = 0.51 + 0.49 \logit^{-1}(\tilde{u}), \quad k = \exp(\kappa), \quad (11)$$

with $\tilde{\ell}, \tilde{u}, \kappa \in \mathbb{R}$. This constrains $\ell \in [0, 0.49]$ and $u \in [0.51, 1]$ into disjoint intervals so that $u > \ell$ by construction, the curve direction is unambiguous, and k is positive on the natural scale.

The inflection point m_{ij} is modelled as a linear function of (mean-centred) assay temperature, with additional fixed covariates and nested random effects:

$$m_{ij} = \beta_0 + \beta_1 (T_{ij} - \bar{T}) + \mathbf{x}_{ij}^\top \gamma + b_j + \varepsilon_{ij}, \quad (12)$$

where $\bar{T}$ is the grand mean of assay temperatures (centring stabilises sampling and decorrelates β_0 from β_1), $\mathbf{x}_{ij}$ holds any additional fixed covariates with coefficients γ , b_j collects the nested random effects and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is an observation-level random effect that absorbs the binomial overdispersion characteristic of replicated count assays (Harrison 2014). However, we could also remove the observation-level random effect from the model and instead fit a beta-binomial error distribution which estimates a dispersion parameter (ϕ) more naturally.

Together, Equation 1–Equation 12 define the hierarchical Bayesian 4PL fit directly to the raw binomial counts. Every classical TDT quantity — slope, z , CTmax at any reference time, and the heat-injury integral — is recovered as a one-line posterior summary of this single joint fit, as derived in §[Equivalence with the classical pipeline].

6 Case Study 1: Thermal sensitivity estimation across life stages in shrimp

6.1 What the 4PL delivers that the two-step cannot

7 Worked example: brown shrimp thermal tolerance

7.1 Data and experimental design

7.2 Lethal TDT — hierarchical 4PL on counts

7.3 Sublethal time-to-death

7.4 Dynamic CTmax across ramping rates

7.5 Reconciling the three endpoints via the Jørgensen ramping equation

7.6 Field-temperature heat-injury projections under warming scenarios

8 Discussion

8.1 What changes for the field if the hierarchical 4PL becomes the default

8.2 Limits of the present analysis

8.3 Recommendations for primary studies and meta-analyses

9 Conclusions

10 Acknowledgements

Author contributions

DN conceived the project, performed the analyses, and wrote the manuscript.

Data and code availability

All data, analysis scripts, and the rendered manuscript are available at <https://github.com/....> The hierarchical 4PL is implemented in [brms](#) (Bürkner 2017) using [cmdstanr](#) as the Stan backend.

Conflict of interest

The authors declare no competing interests.

References

- Arnold, P.A., Noble, D.W.A., Nicotra, A.B., Kearney, M.R., Rezende, E.L., Andrew, S.C., *et al.* (2025). [A framework for modelling thermal load sensitivity across life](#). *Global Change Biology*, 31, e70315.
- Bolker, B.M. (2008). *Ecological models and data in R*. Princeton University Press, Princeton, NJ.
- Buckley, L.B., Huey, R.B. & Ma, C.-S. (2025). [How damage, recovery, and repair alter the fitness impacts of thermal stress](#). *Integrative and Comparative Biology*, 65, 1061–1075.
- Burke, D.L., Ensor, J. & Riley, R.D. (2017). [Meta-analysis using individual participant data: One-stage and two-stage approaches, and why they may differ](#). *Statistics in Medicine*, 36, 855–875.
- Bürkner, P.-C. (2017). [brms: An R package for Bayesian multilevel models using Stan](#). *Journal of Statistical Software*, 80, 1–28.
- Carroll, R.J., Ruppert, D., Stefanski, L.A. & Crainiceanu, C.M. (2006). [Measurement error in nonlinear models: A modern perspective](#). 2nd edn. Chapman and Hall/CRC.
- Deutsch, C.A., Tewksbury, J.J., Huey, R.B., Sheldon, K.S., Ghalambor, C.K., Haak, D.C., *et al.* (2008). [Impacts of climate warming on terrestrial ectotherms across latitude](#). *Proceedings of the National Academy of Sciences*, 105, 6668–6672.
- Gelman, A. (2006). [Prior distributions for variance parameters in hierarchical models \(comment on article by Browne and Draper\)](#). *Bayesian Analysis*, 1, 515–534.
- Harrison, X.A. (2014). [Using observation-level random effects to model overdispersion in count data in ecology and evolution](#). *PeerJ*, 2, e616.
- Harrison, X.A., Donaldson, L., Correa-Cano, M.E., Evans, J., Fisher, D.N., Goodwin, C.E.D., *et al.* (2018). [A brief introduction to mixed effects modelling and multi-model inference in ecology](#). *PeerJ*, 6, e4794.
- Jørgensen, L.B., Malte, H. & Overgaard, J. (2019). [How to assess Drosophila heat tolerance: Unifying static and dynamic tolerance assays to predict heat distribution limits](#). *Functional Ecology*, 33, 629–642.
- Jørgensen, L.B., Malte, H. & Overgaard, J. (2021). [A unifying model to estimate thermal tolerance limits in ectotherms across static, dynamic and fluctuating exposures to thermal stress](#). *Scientific Reports*, 11, 12840.
- Jørgensen, L.B., Ørsted, M., Malte, H., Wang, T. & Overgaard, J. (2022). [Extreme escalation of heat failure rates in ectotherms with global warming](#). *Nature*, 611, 93–98.
- Kingsolver, J.G. & Umbanhowar, J. (2018). [The analysis and interpretation of critical temperatures](#). *Journal of Experimental Biology*, 221, jeb167858.
- Li, Y.-J., Chen, S.-Y., Jørgensen, L.B., Overgaard, J., Renault, D., Colinet, H., *et al.* (2023). [Interspecific differences in thermal tolerance landscape explain aphid community abundance under climate change](#). *Journal of Thermal Biology*, 114, 103583.
- Nakagawa, S. & Santos, E.S.A. (2012). [Methodological issues and advances in biological meta-analysis](#). *Evolutionary Ecology*, 26, 1253–1274.
- Nielsen, O.K., Ritz, C. & Streibig, J.C. (2004). [Nonlinear mixed-model regression to analyze herbicide dose-response relationships](#). *Weed Technology*, 18, 30–37.

- Noble, D.W.A., Mayfield, M.M., Hoffmann, A.A., Chen, Z.-H., Lade, S.J., Bai, X., *et al.* (2026). [A systems modelling approach to predict biological responses to extreme heat](#). *Trends in Ecology & Evolution*, 41, 451–464.
- Ørsted, M., Jørgensen, L.B. & Overgaard, J. (2022). [Finding the right thermal limit: A framework to reconcile ecological, physiological and methodological aspects of CTmax in ectotherms](#). *Journal of Experimental Biology*, 225, jeb244514.
- Ørsted, M., Willot, Q., Olsen, A.K., Kongsgaard, V. & Overgaard, J. (2024). [Thermal limits of survival and reproduction depend on stress duration: A case study of *Drosophila suzukii*](#). *Ecology Letters*, 27, e14421.
- Pinsky, M.L., Eikeset, A.M., McCauley, D.J., Payne, J.L. & Sunday, J.M. (2019). [Greater vulnerability to warming of marine versus terrestrial ectotherms](#). *Nature*, 569, 108–111.
- Pottier, P., Kearney, M.R., Wu, N.C., Gunderson, A.R., Rej, J.E., Rivera-Villanueva, A.N., *et al.* (2025). [Vulnerability of amphibians to global warming](#). *Nature*, 639, 954–961.
- Rezende, E.L., Bozinovic, F., Szilágyi, A. & Santos, M. (2020). [Predicting temperature mortality and selection in natural *Drosophila* populations](#). *Science*, 369, 1242–1245.
- Rezende, E.L., Castañeda, L.E. & Santos, M. (2014). [Tolerance landscapes in thermal ecology](#). *Functional Ecology*, 28, 799–809.
- Ritz, C. (2010). [Toward a unified approach to dose-response modeling in ecotoxicology](#). *Environmental Toxicology and Chemistry*, 29, 220–229.
- Ritz, C., Baty, F., Streibig, J.C. & Gerhard, D. (2015). [Dose-response analysis using R](#). *PLoS ONE*, 10, e0146021.
- Ritz, C. & Streibig, J.C. (2008). *Nonlinear regression with R*. Springer-Verlag, New York.
- Rudemo, M., Ruppert, D. & Streibig, J.C. (1989). [Random-effect models in nonlinear regression with applications to bioassay](#). *Biometrics. Journal of the International Biometric Society*, 45, 349–362.
- Schielzeth, H. (2010). [Simple means to improve the interpretability of regression coefficients](#). *Methods in Ecology and Evolution*, 1, 103–113.
- Smale, D.A., Wernberg, T., Oliver, E.C.J., Thomsen, M., Harvey, B.P., Straub, S.C., *et al.* (2019). [Marine heatwaves threaten global biodiversity and the provision of ecosystem services](#). *Nature Climate Change*, 9, 306–312.
- Snook, R.R., Bretman, A., Dougherty, L.R. & Fricke, C. (2026). [The consequences of rising temperatures for animal fertility](#). *Nature Reviews Biodiversity*, 2, 225–241.
- Sunday, J.M., Bates, A.E. & Dulvy, N.K. (2011). [Global analysis of thermal tolerance and latitude in ectotherms](#). *Proceedings of the Royal Society B: Biological Sciences*, 278, 1823–1830.
- Sunday, J.M., Bates, A.E. & Dulvy, N.K. (2012). [Thermal tolerance and the global redistribution of animals](#). *Nature Climate Change*, 2, 686–690.
- Truebano, M., Fenner, P., Tills, O., Rundle, S. & Rezende, E. (2018). [Thermal strategies vary with life history stage](#). *Journal of Experimental Biology*, 221, jeb171629.
- Villeneuve, A.R. & White, E.R. (2024). [Predicting organismal response to marine heatwaves using dynamic thermal tolerance landscape models](#). *Journal of Animal Ecology*, 94, 1515–1527.
- Wiens, J.J. (2016). [Climate-related local extinctions are already widespread among plant and animal species](#). *PLOS Biology*, 14, e2001104.

11 A Tutorial with Simulations

This section is a guided tutorial using simulated data. We show, step by step, how the Bayesian hierarchical 4PL — under both binomial sampling and the more realistic beta-binomial (overdispersed) case — can recover thermal sensitivity (z) and $CT_{max_{1hr}}$ from the single joint model, yet provide substantial downstream benefits in terms of propagating uncertainty and giving more flexibility. We then complement the simulation with a few case studies which walk through real data that is presented within the main manuscript in other sections.

11.0.1 Simulating Data

11.0.1.1 Step 1 — Setting up parameters

We choose values that might be typical of a thermal mortality assay. The four 4PL parameters are the lower and upper asymptotes (ℓ, u), the slope on the $\log_{10}$ -time axis (k), and a midpoint that varies linearly with temperature: $\text{mid}(T) = \beta_0 + \beta_1(T - \bar{T})$. Each value below has a clear biological meaning.

```
true_params <- list(  
  ell      = 0.03,    # 3% of individuals survive even at the longest exposures  
  u        = 0.97,    # 97% are alive at very short exposures  
  k        = 8,       # steepness of the survival drop on the log10(t) axis  
  m_beta0  = 1.5,     # midpoint at the grand-mean temperature: ~31.6 min  
  m_beta1  = -0.15,   # midpoint drops 0.15 log10-min per °C; z = -1/beta1  6.7 °C  
  T_bar    = 34       # grand-mean temperature  
)
```

The implied thermal sensitivity is $z = -1/\beta_1 \approx 6.7^\circ\text{C}$ — the temperature change that shifts LT50 by a factor of 10. From the same parameters we can also compute the simulation truth for the uncentred TDT intercept $\alpha = \beta_0 + \frac{1}{k} \log \frac{u-0.5}{0.5-\ell} - \beta_1 \bar{T}$ and the critical thermal limit at 1 hour $CT_{max_{1hr}} = (\log_{10} 60 - \alpha)/\beta_1$ — both approaches below will be checked against these known values.

Recall that we can calculate ℓ , z and $CT_{max_{1hr}}$ from the true parameters using the equations in the main manuscript. We'll do that now so we have a comparator for later.

```
true_z      <- -1 / true_params$m_beta1  
true_alpha  <- true_params$m_beta0 +  
  (1 / true_params$k) *  
  log((true_params$u - 0.5) / (0.5 - true_params$ell)) -  
  true_params$m_beta1 * true_params$T_bar  
true_CTmax  <- (log10(60) - true_alpha) / true_params$m_beta1
```

11.0.1.2 Step 2 — Simulated study design

We mimic a textbook TDT experiment: a grid of five assay temperatures crossed with six log-spaced exposure durations, repeated 30 times per cell, with 30 individuals tested per replicate. That gives 900 trials and 27,000 individuals — large enough to recover the parameters with negligible uncertainty, but small enough to fit in a few minutes. Of course, in practice we will not have this much data as will be seen in the case studies, but simulations are a good starting point to understand the models being fit and the conversions.

```
temps      <- c(30, 32, 34, 36, 38)          # 5 assay temperatures (°C)
durations  <- c(1, 5, 15, 45, 135, 405)      # 6 durations (minutes, log-spaced)
n_rep      <- 30                             # 30 replicates per cell
n_per_rep  <- 30                             # 30 individuals per replicate

design <- tidyr::expand_grid(
  T      = temps,
  t      = durations,
  rep    = seq_len(n_rep)
) |>
dplyr::mutate(
  log10_t = log10(t),
  T_c     = T - true_params$T_bar,           # mean-centred T
  mid     = true_params$m_beta0 + true_params$m_beta1 * T_c, # mid(T)
  p_true  = true_params$ell +
    (true_params$u - true_params$ell) /
    (1 + exp(true_params$k * (log10_t - mid))),
  n       = n_per_rep
)
```

11.0.1.3 Step 3 — Look at the truth before simulating

Before generating noisy data, let's plot the *true* survival surface (Figure S1) so we know what the model is being asked to recover. Each line is the 4PL at one assay temperature, plotted against $\log_{10}$ exposure time.

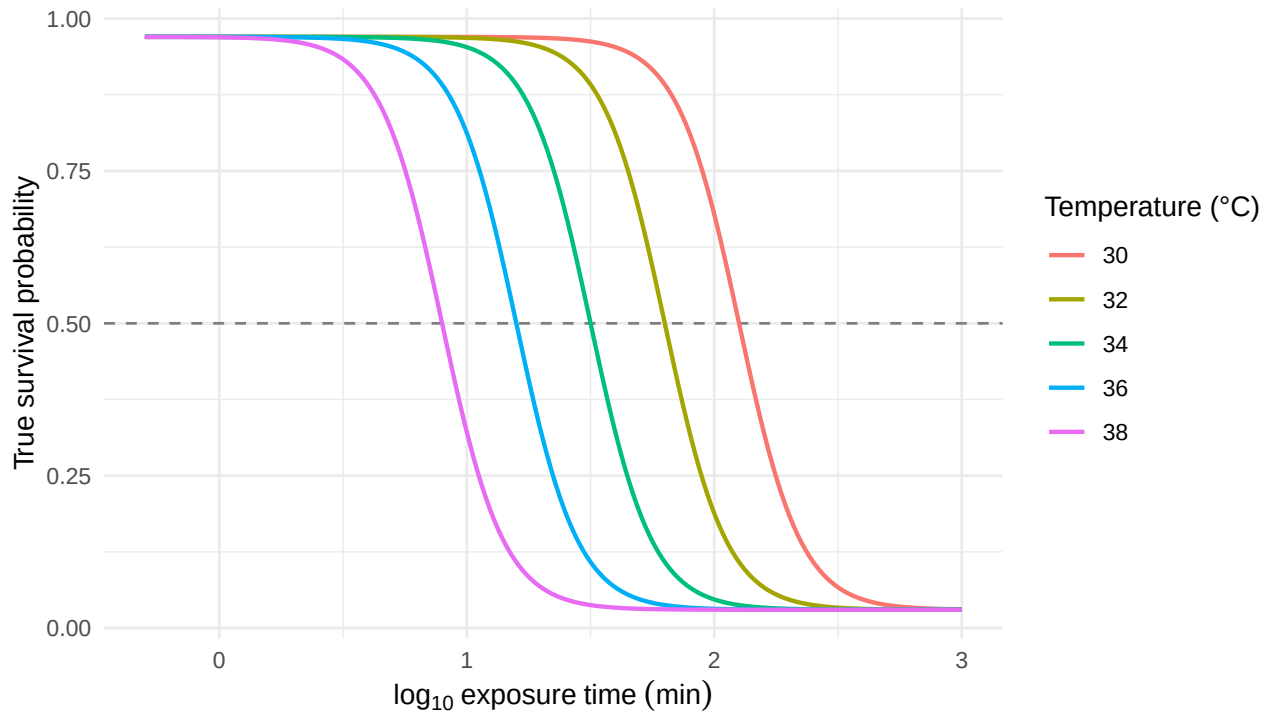


Figure S1: The known 4PL dose-response surface used to generate the simulated data, evaluated on a dense time grid at each of the five assay temperatures. The horizontal line marks 50% survival; where each curve crosses it gives the true LT50 at that temperature.

Notice the curves shift left as temperature rises: hotter temperatures kill faster, so the LT50 is reached at shorter durations. That horizontal shift in midpoints is exactly what β_1 encodes.

11.0.1.4 Step 4 — Generate noisy data: binomial and beta-binomial

Real survival counts have at least two flavours of noise. **Binomial** sampling is the textbook case: each replicate's survival count is a draw from $\text{Binomial}(n, p)$ with p from the 4PL surface. **Beta-binomial** sampling adds *overdispersion* — replicate-to-replicate variability in p that the binomial cannot capture, e.g., from cohort effects or unmodelled heterogeneity. Real TDT data almost always show overdispersion so in many cases the beta binomial will be a better fit.

```
# (a) Binomial: y ~ Binomial(n, p_true)
sim_bin <- design |>
  dplyr::mutate(y = rbinom(dplyr::n(), size = n, prob = p_true))

# (b) Beta-binomial: replicate-level p drawn from Beta(p*phi, (1-p)*phi)
phi <- 5 # smaller phi = more overdispersion
sim_bb <- design |>
  dplyr::mutate(
    p_draw = rbeta(dplyr::n(), p_true * phi, (1 - p_true) * phi),
    y       = rbinom(dplyr::n(), size = n, prob = p_draw)
  )
```

Table S1: Per-temperature Stage-1 estimates of $\log_{10}$ LT50 from a binomial GLM with logit link, fit separately at each of the five assay temperatures. Truth values are the simulated midpoints (which coincide with $\log_{10}$ LT50 here because the simulation asymptotes are near 0 and 1).

T (°C)	Truth $\log_{10}(\text{LT50})$	Binomial $\log_{10}(\text{LT50})$	Beta-binomial $\log_{10}(\text{LT50})$
30	2.1	2.061	2.082
32	1.8	1.795	1.8
34	1.5	1.48	1.528
36	1.2	1.202	1.195
38	0.9	0.907	0.954

11.0.2 The classical two-stage TDT pipeline

We first run the simulated data through the **classical two-stage TDT pipeline** (Ørsted *et al.* 2022; Rezende *et al.* 2014), which is dominant in the thermal-tolerance literature.

1. **Stage 1.** At each assay temperature, fit a logistic dose-response curve to the count data and read off a point estimate of $\log_{10}$ LT50_T.
2. **Stage 2.** Regress those Stage-1 point estimates on assay temperature with ordinary least squares, then derive $z = -1/\beta_1$ and $CT_{max_{1hr}} = (\log_{10} 60 - \alpha)/\beta_1$ from the fitted line.

11.0.2.1 Stage 1 — Dose-response curves at each temperature

We fit a binomial GLM with a logit link separately at each assay temperature. This is the most-common Stage-1 specification in the TDT literature: a two-parameter logistic with asymptotes constrained to 0% and 100%. The midpoint $\log_{10}$ LT50_T = $-\beta_0/\beta_1$ falls out of the GLM’s two coefficients.

```
fit_stage1 <- function(sim_data) {
  sim_data |>
    dplyr::group_by(T) |>
    dplyr::group_modify(function(d, key) {
      f <- glm(cbind(y, n - y) ~ log10_t, data = d, family = binomial())
      co <- coef(f)
      tibble::tibble(log10_lt50 = as.numeric(-co[1] / co[2]))
    }) |>
    dplyr::ungroup()
}

stage1_bin <- fit_stage1(sim_bin)
stage1_bb <- fit_stage1(sim_bb)
```

With the simulation’s true asymptotes near 0 and 1, the 2PL’s forced-asymptote constraint introduces negligible bias here, so each Stage-1 estimate sits close to the simulation truth.

11.0.2.2 Stage 2 — log-linear TDT regression

Stage 2 regresses Stage-1 $\log_{10}$ LT50 on assay temperature with ordinary least squares. The classical TDT quantities — $z = -1/\beta_1$ and $CT_{max_{1hr}} = (\log_{10} 60 - \alpha)/\beta_1$ — fall out of the fitted line.

```
fit_stage2 <- function(stage1) {  
  fit2 <- lm(log10_lt50 ~ T, data = stage1)  
  co    <- coef(fit2)  
  
  tibble::tibble(  
    quantity = c("z (°C)", "CTmax at 1 hr (°C)"),  
    est      = c(as.numeric(-1 / co["T"]),  
                  as.numeric((log10(60) - co["(Intercept)"]) / co["T"]))  
  )  
}  
  
ts_bin <- fit_stage2(stage1_bin)  
ts_bb  <- fit_stage2(stage1_bb)
```

Figure S2 visualises the Stage-2 fit. Each point is one Stage-1 $\log_{10}$ LT50 estimate from Table S1; the solid line is the OLS regression through them. The slope is β_1 , from which $z = -1/\beta_1$ — i.e., the temperature increase that causes LT50 to change by a factor of 10. The line's height at $\log_{10} 60 \approx 1.78$ (1 hour) is at the temperature equal to $CT_{max_{1hr}}$ — the vertical dotted line drops from this intersection straight to the temperature axis to mark it. The dashed grey line is the simulation truth.

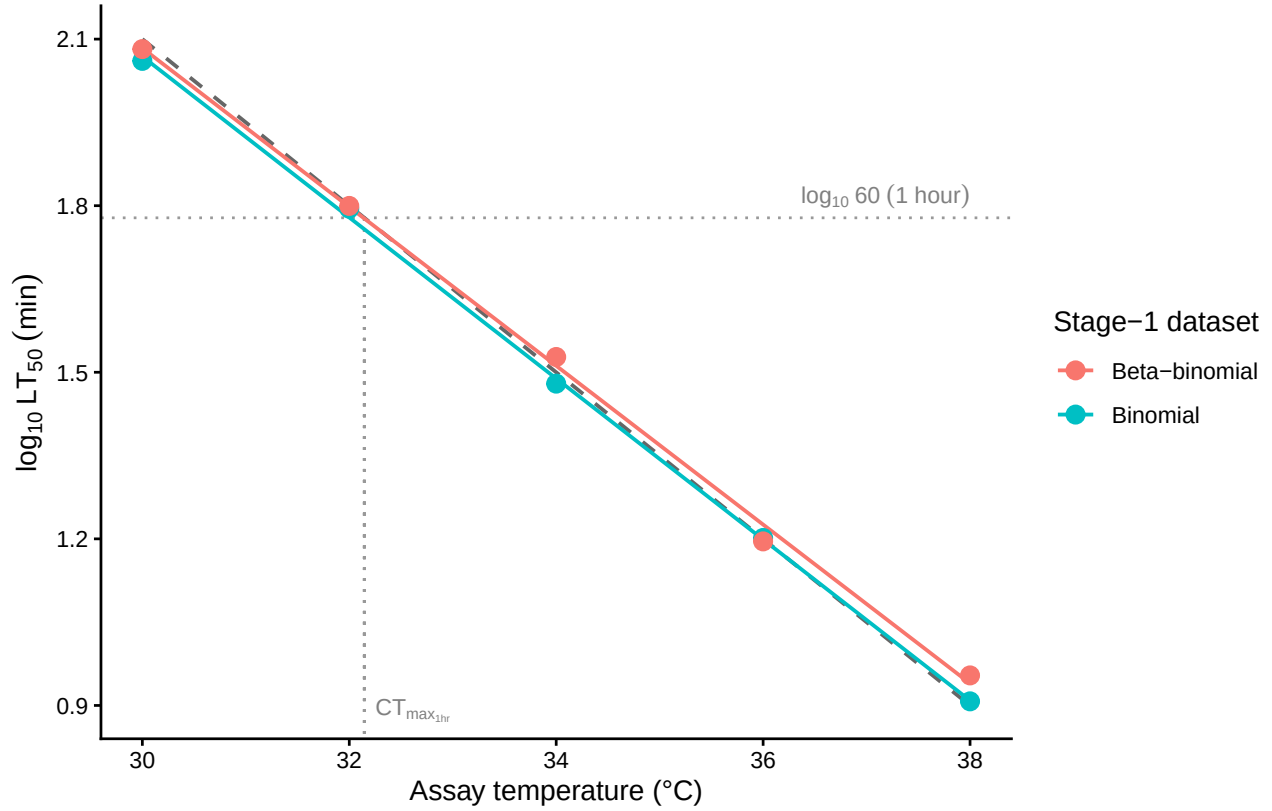


Figure S2: Stage-2 OLS regression of $\log_{10}$ LT50 on assay temperature, fit separately to the binomial and beta-binomial simulated datasets. Points are the Stage-1 per-temperature LT50 estimates; solid coloured lines are the OLS fits; the dashed grey line is the simulation truth. The slope of each line equals β_1 , from which $z = -1/\beta_1$. The horizontal dotted line marks $\log_{10} 60$ (1 hour); the vertical dotted line drops from where the truth crosses 1 hour straight down to the temperature axis, marking $CT_{max_{1hr}}$ visually.

This is the entire two-stage workflow. The point estimates that come out — z and $CT_{max_{1hr}}$ — are what the field reports. We carry them forward (`ts_bin` and `ts_bb`) for the side-by-side comparison once we have the joint Bayesian results.

! Important

Two-stage steps often ignore uncertainty in estimates in both fitting and error propagation. Uncertainty in estimates can be propagated using the SE's and the delta method, but only at each individual stage.

11.0.3 A joint Bayesian 4PL

Instead of fitting a separate logistic at each assay temperature and then regressing the LT50s, the joint fit estimates every parameter — ℓ , u , k , β_0 and β_1 — in a single hierarchical model on the raw counts. We now use the same simulated data to derive the z and $CT_{max_{1hr}}$ from the posterior, in one model rather than two.

11.0.3.1 Specifying the 4PL in brms

The model has two pieces. The first is the 4PL itself — survival probability as a function of $\log_{10}$ exposure time:

$$p_{ij} = \ell + \frac{u - \ell}{1 + \exp(k(\log_{10} t_{ij} - \text{mid}_{ij}))}, \quad (\text{S1})$$

where ℓ and u are the lower and upper asymptotes, k is the slope on the $\log_{10} t$ axis, and mid_{ij} is the value of $\log_{10} t$ at which the curve crosses the midpoint between ℓ and u . The second piece lets the midpoint vary linearly with mean-centred temperature:

$$\text{mid}_{ij} = \beta_0 + \beta_1 (T_{ij} - \bar{T}), \quad (\text{S2})$$

so that β_1 encodes how fast the dose-response curve shifts on the time axis as assay temperature changes; $\bar{T}$ is the grand-mean temperature, included so that β_0 is interpretable as the midpoint at the average assay temperature.

In **brms** non-linear syntax these two pieces become one formula and four sub-formulas — one for each parameter in the 4PL model:

```
# Soft constraints to keep the parameters in their natural ranges.
# These are not informative priors - just the bounds the parameters
# must obey to be biologically meaningful.
priors_4pl <- c(
  prior(beta(1, 5), nlpar = "ell", lb = 0, ub = 1),
  prior(beta(5, 1), nlpar = "u", lb = 0, ub = 1),
  prior(normal(0, 10), nlpar = "k", lb = 0),
  prior(normal(0, 5), nlpar = "mid")
)

bform_bin <- bf(
  y | trials(n) ~ ell + (u - ell) / (1 + exp(k * (log10_t - mid))),
  ell ~ 1, u ~ 1, k ~ 1, mid ~ T_c,
  nl = TRUE,
  family = binomial(link = "identity")
)

# Same model, beta-binomial likelihood for the overdispersed dataset.
bform_bb <- bf(
  y | trials(n) ~ ell + (u - ell) / (1 + exp(k * (log10_t - mid))),
  ell ~ 1, u ~ 1, k ~ 1, mid ~ T_c,
  nl = TRUE,
  family = beta_binomial(link = "identity")
)
```

The first line in **bf()** is Equation S1. The next four lines say (i) the asymptotes and slope are simply scalars and we estimate an overall intercept for each ($\text{ell} \sim 1, u \sim 1, k \sim 1$) and (ii) the midpoint

depends linearly on mean-centred temperature ($\text{mid} \sim T_c$), recovering Equation S2. This is the entire model.

! Important

The model we fit is rather simple — it can be extended in several ways. For example, random effects can be added to any of the four 4PL parameters; for instance, $\text{mid} \sim T_c + (1 \mid \text{day})$ allows the midpoint to vary across assay days or batches. Temperature can also be a predictor for the other 4PL parameters — adding $k \sim T_c$, for instance, would let the slope of the dose-response curve itself depend on temperature. Random effects in the joint fit also enable a much more detailed variance decomposition than the two-stage pipeline can deliver. We'll explore more of this in Section 11.0.6.

11.0.3.2 Fitting the model

Now that we have the model formula set up and priors we can fit the model. We pass `file = ...` and `file_refit = "on_change"` directly to `brm()`. This tells `brms` to save the fit to disk on first run and to reload it on subsequent runs unless the model or data has changed.

```
# Disable Quarto chunk caching: brms's `file = ...` / `file_refit = "on_change"`
# already handles refitting on formula/data changes. Quarto's own cache would
# hold a stale fit object across renders.

fit_bin <- brm(
  bform_bin, data = sim_bin, prior = priors_4pl,
  chains = 4, iter = 2000, cores = 4, seed = 123,
  file = file.path(models_dir, "sim_4pl_binomial"),
  file_refit = "on_change"
)

fit_bb <- brm(
  bform_bb, data = sim_bb, prior = priors_4pl,
  chains = 4, iter = 2000, cores = 4, seed = 123,
  file = file.path(models_dir, "sim_4pl_betabinomial"),
  file_refit = "on_change"
)
```

We can also get an idea of how much variation our model explains that considers all sources of relevant variation.

```
# Obtain the Bayes R2 for each model
brms::bayes_R2(fit_bin)
```

	Estimate	Est.Error	Q2.5	Q97.5
R2	0.9875292	0.0001264011	0.987227	0.9877159

Table S2: Posterior medians and 95% credible intervals for the four 4PL parameters and the temperature slope, compared to the known truth used to generate the simulated data. Recovery is judged successful when the credible interval covers the truth.

Parameter	Truth	Bin. median	Bin. lower	Bin. upper	BB median	BB lower	BB upper
ell	0.03	0.0288	0.0252	0.0329	0.0295	0.0223	0.0381
u	0.97	0.9675	0.9637	0.9709	0.9679	0.9598	0.9743
k	8	8.3196	7.9162	8.7628	7.5159	6.8119	8.3283
mid (intercept)	1.5	1.5012	1.4927	1.5095	1.5157	1.4951	1.5366
mid (slope)	-0.15	-0.15	-0.1527	-0.1471	-0.1474	-0.1544	-0.1404

```
brms::bayes_R2(fit_bb)
```

```

      Estimate Est.Error      Q2.5      Q97.5
R2 0.9263016 0.00105021 0.9238746 0.9280369

```

11.0.3.3 Checking parameter recovery

We can now check how well our fitted model recovered the true parameters that generated the data. The most direct check is to compare posterior medians (with 95% credible intervals) to the known truth (Table S2). We have big sample sizes here so our values should be pretty close to our true values.

We can also overlay the posterior fitted curves on the simulated data (Figure S3). If the model has recovered the true surface, the fitted lines should track the simulated proportions at every assay temperature.

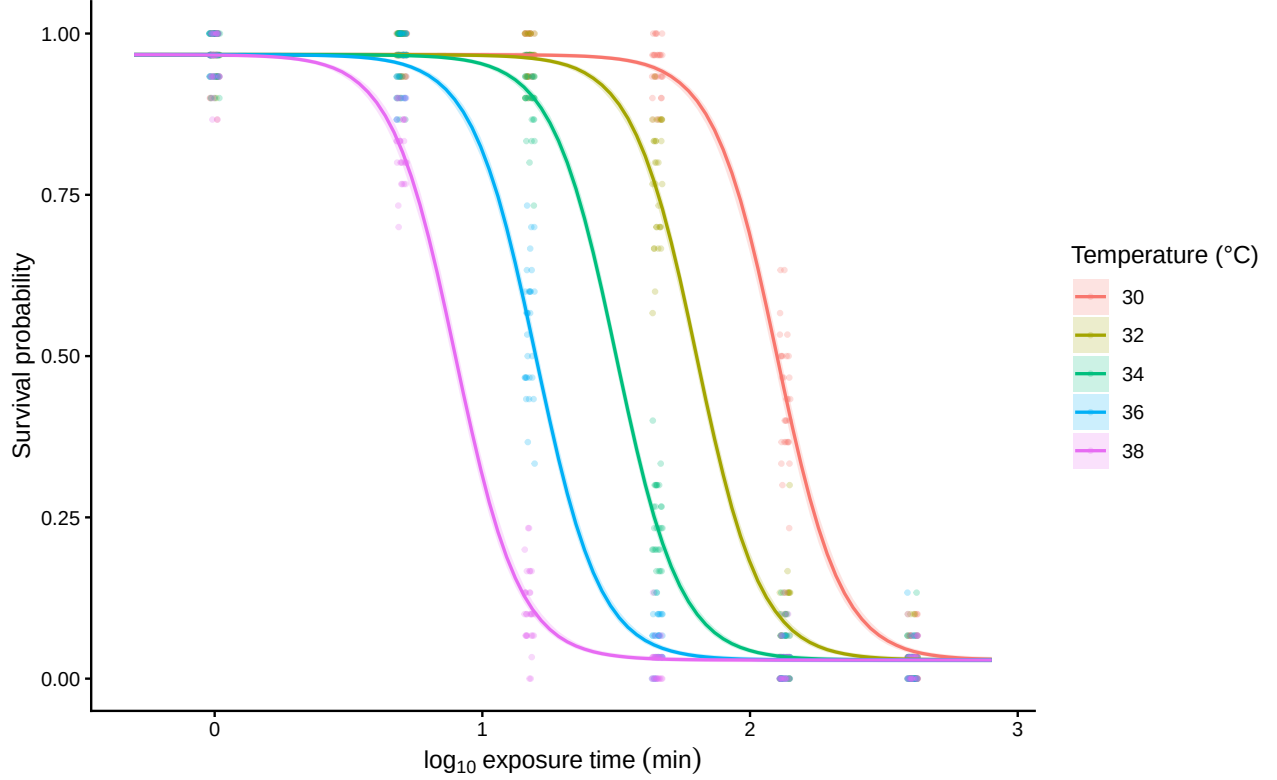


Figure S3: Posterior fitted 4PL curves (solid lines, with shaded 95% credible bands) overlaid on the simulated binomial data (points). One panel-coloured line per assay temperature. Where the fitted curves track the simulated proportions across all five temperatures, the model has recovered the underlying dose-response surface.

11.0.4 Deriving z and $CT_{max_{1hr}}$ from the posterior

The whole point of fitting the joint Bayesian 4PL is that the classical TLS quantities — thermal sensitivity (z) and the critical thermal limit at a reference duration ($CT_{max_{1hr}}$) — fall out as one-line transformations of the posterior we just fit. We never refit the model; we just push the posterior draws of $(\ell, u, k, \beta_0, \beta_1)$ through the same closed-form expressions used to derive the simulation truth in Step 1.

```
# Helper: push posterior draws of (ell, u, k, beta0, beta1) through the closed-
# form derivations from the main text to get z and CTmax at 1 hour.
derive_z_ctmax <- function(fit) {
  posterior::as_draws_df(fit) |>
    dplyr::mutate(
      z = -1 / b_mid_T_c,
      alpha = b_mid_Intercept +
        (1 / b_k_Intercept) *
        log((b_u_Intercept - 0.5) / (0.5 - b_ell_Intercept)) -
        b_mid_T_c * true_params$T_bar,
      CTmax_1hr = (log10(60) - alpha) / b_mid_T_c
    )
}
```

Table S3: Posterior medians and 95% credible intervals for thermal sensitivity (z) and the critical thermal limit at a 1-hour reference duration ($CT_{max_{1hr}}$), derived directly from the joint Bayesian 4PL fit. Both quantities recover the simulation truth under both binomial and beta-binomial likelihoods.

Quantity	Truth	Bin. median	Bin. lower	Bin. upper	BB median	BB lower	BB upper
z (°C)	6.67	6.67	6.55	6.8	6.78	6.47	7.12
CTmax at 1 hr (°C)	32.15	32.15	32.08	32.2	32.21	32.06	32.37

```

    ) |>
    dplyr::select(z, CTmax_1hr)
  }

# Apply to both fits to get posterior draws of (z, CTmax_1hr).
draws_bin_d <- derive_z_ctmax(fit_bin)
draws_bb_d  <- derive_z_ctmax(fit_bb)

```

- ① Pull the joint posterior of $(\ell, u, k, \beta_0, \beta_1)$ from the brms fit as a data frame, with one row per posterior draw.
- ② For each draw, compute thermal sensitivity $z = -1/\beta_1$.
- ③ Compute the asymmetry-corrected intercept $\alpha = \beta_0 + (1/k) \log((u - 0.5)/(0.5 - \ell)) - \beta_1 \bar{T}$ used in the closed-form CT_{max} expression.
- ④ Compute $CT_{max_{1hr}} = (\log_{10} 60 - \alpha)/\beta_1$, the temperature at which median lethal time equals 1 hour.
- ⑤ Keep only the two derived quantities — the posterior is now a draws-by-2 tibble.

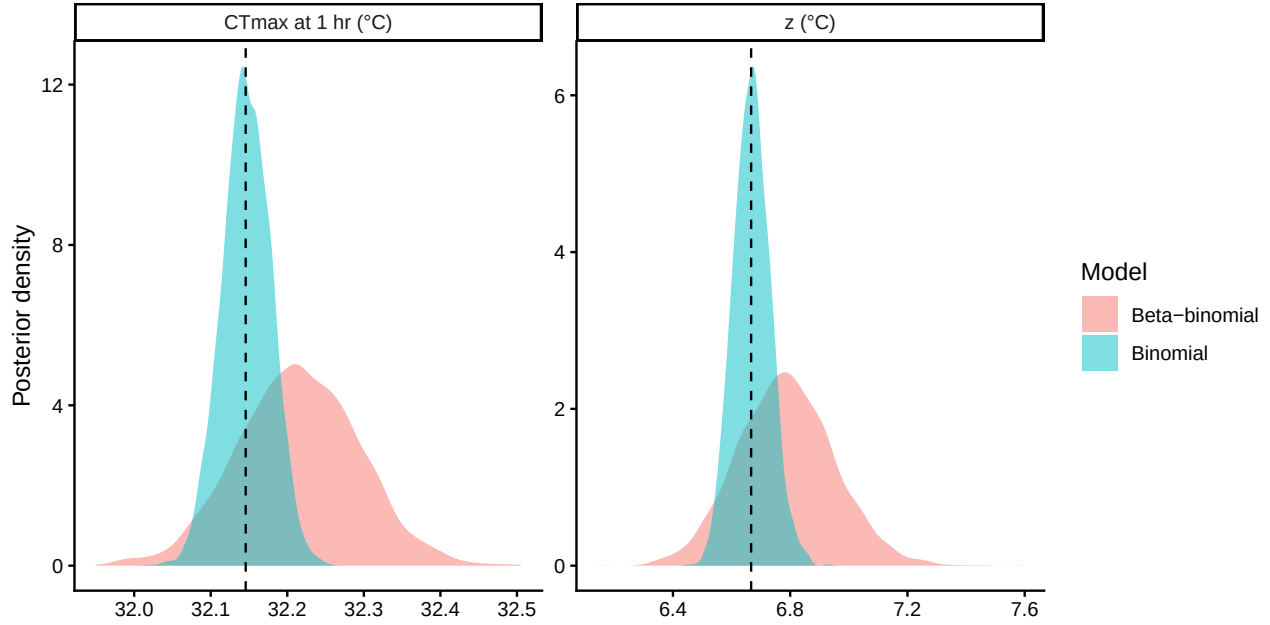


Figure S4: Posterior distributions for thermal sensitivity (z) and $CT_{max_{1hr}}$ from the binomial fit (blue) and beta-binomial fit (orange). Vertical dashed lines mark the simulation truth. Both posteriors are tightly concentrated around the truth, illustrating that the joint Bayesian fit recovers the classical TLS quantities with calibrated uncertainty without any additional regression step.

The two derived quantities recover their simulation truth in both fits (Table S3, Figure S4), with credible intervals that reflect joint uncertainty in all four 4PL parameters and the temperature slope. Crucially, **we did not run a second regression** to get z and $CT_{max_{1hr}}$. They are properties of the same posterior we already have. This is the key practical advantage of the joint approach over the two-step pipeline: every classical TLS quantity is one transformation away from the fit we have already run, and every quantity inherits the same calibrated uncertainty.

Given that we have the full distribution of z and $CT_{max_{1hr}}$, we can now report on uncertainty in their values *very* easily. All we need to do is take the standard deviation of this distribution for the standard error or we can even summarise with quantiles as follows:

```
summarise_draws <- function(x) {
  q <- quantile(x, c(0.025, 0.5, 0.975), names = FALSE)
  tibble::tibble(
    mean    = mean(x),
    sd      = sd(x),
    median  = q[2],
    lower   = q[1],
    upper   = q[3]
  )
}

uncert_summary <- dplyr::bind_rows(
  summarise_draws(draws_bin_d$z) |>
```

Table S4: Posterior summaries of z and $CT_{max_{1hr}}$ from the joint Bayesian 4PL: posterior mean, standard deviation, median, and 95% credible interval (2.5% and 97.5% quantiles), reported separately for the binomial and beta-binomial fits.

Quantity	Likelihood	Mean	SD	Median	Lower	Upper
z (°C)	Binomial	6.67	0.0629	6.67	6.55	6.8
CTmax at 1 hr (°C)	Binomial	32.15	0.032	32.15	32.08	32.21
z (°C)	Beta-binomial	6.79	0.1643	6.78	6.47	7.12
CTmax at 1 hr (°C)	Beta-binomial	32.21	0.0793	32.21	32.06	32.37

```

dplyr::mutate(quantity = "z (°C)", likelihood = "Binomial"),
summarise_draws(draws_bin_d$CTmax_1hr) |>
dplyr::mutate(quantity = "CTmax at 1 hr (°C)", likelihood = "Binomial"),
summarise_draws(draws_bb_d$z) |>
dplyr::mutate(quantity = "z (°C)", likelihood = "Beta-binomial"),
summarise_draws(draws_bb_d$CTmax_1hr) |>
dplyr::mutate(quantity = "CTmax at 1 hr (°C)", likelihood = "Beta-binomial")
) |>
dplyr::select(quantity, likelihood, mean, sd, median, lower, upper)

# Scalars for inline reporting in the next paragraph.
sd_z_bin <- round(sd(draws_bin_d$z), 2)
sd_z_bb <- round(sd(draws_bb_d$z), 2)
sd_ct_bin <- round(sd(draws_bin_d$CTmax_1hr), 2)
sd_ct_bb <- round(sd(draws_bb_d$CTmax_1hr), 2)
ci_z_bin <- round(diff(quantile(draws_bin_d$z, c(0.025, 0.975), names = FALSE)), 2)
ci_z_bb <- round(diff(quantile(draws_bb_d$z, c(0.025, 0.975), names = FALSE)), 2)
ci_ct_bin <- round(diff(quantile(draws_bin_d$CTmax_1hr, c(0.025, 0.975), names = FALSE)), 2)
ci_ct_bb <- round(diff(quantile(draws_bb_d$CTmax_1hr, c(0.025, 0.975), names = FALSE)), 2)

uncert_summary |>
dplyr::rename(
  Quantity = quantity,
  Likelihood = likelihood,
  Mean = mean,
  SD = sd,
  Median = median,
  Lower = lower,
  Upper = upper
) |>
tinytable::tt(digits = 3)

```

In short, both z and $CT_{max_{1hr}}$ are estimated with high precision and their credible intervals come straight from the joint posterior. If we fit this model using frequentist approaches we could do the same and use the delta-method, or bootstrapping to get uncertainty.

Table S5: Point estimates (two-stage) and posterior medians (joint Bayesian) for thermal sensitivity (z) and $CT_{max_{1hr}}$, recovered by each of four estimator–dataset combinations from the same simulated datasets. Truth values are the simulation parameters.

Quantity	Truth	Two-stage (binomial)	Joint Bayesian (binomial)	Two-stage (beta-binomial)
z (°C)	6.67	6.9	6.67	6.99
CTmax at 1 hr (°C)	32.15	32	32.15	32.14

Under the binomial likelihood, the standard error (or SD of posterior) for z is 0.06 °C with a 95% credible interval 0.25 °C wide; $CT_{max_{1hr}}$ has SD 0.03 °C and a 95% credible interval width of 0.12 °C.

The beta-binomial fit is slightly wider on both quantities (SD on z : 0.16 °C; on $CT_{max_{1hr}}$: 0.08 °C) because ϕ has absorbed the extra-binomial variance and that uncertainty propagates honestly into the derived quantities — none of which falls out of the two-stage pipeline without an additional bootstrap or delta-method step.

! Important

It is always important to evaluate whether a binomial and a beta-binomial model better fits real data. The extra dispersion, typical of real biological data, should be estimated when it's present as it has important consequences for inferences.

11.0.5 Comparing the two pipelines

11.0.5.1 Point-estimate agreement on the same data

Table S5 puts the two-stage point estimates next to the joint Bayesian posterior medians for both simulated datasets.

All four estimators land on essentially the same point estimates for z and $CT_{max_{1hr}}$, though we notice the two stage is biased upwards slightly, and all sit close to the simulation truth. The joint Bayesian 4PL gives the same answer the classical pipeline gives — it is *not* in disagreement with the field's existing point estimates, it reproduces them.

11.0.6 Extending the model: temperature-varying shape parameters

The closed-form derivations of z and $CT_{max_{1hr}}$ assumed that ℓ, u and k are scalars — they do not depend on T , so the asymmetry correction $\frac{1}{k} \log \frac{u-0.5}{0.5-\ell}$ is a single number rather than a function of temperature. This is convenient but not required. If ℓ, u and k also vary with temperature, then we need to predict how each of these parameters changes with temperature and apply a temperature-varying correction factor. Linear effects of temperature on the shape parameters of the dose-response curve are mechanistically interpretable and biologically plausible. Three concrete examples come to mind:

- **u ~ T_c** lets the upper asymptote (baseline survival at very short exposures) decline with rising assay temperature. This may be quite a sensible assumption if brief exposures at high T cause some immediate mortality.
- **ell ~ T_c** lets the residual surviving fraction at very long exposures depend on T . While this seems less plausible, if there is individual variation (which seems likely) some fraction may still survive, particularly at less extreme temperatures, which could shift the lower baseline across each temperature treatment.
- **k ~ T_c** lets the dose-response steepness depend on T . Biologically this is the most interesting case: protein denaturation kinetics often look more threshold-like (sharper k) near a protein's melting temperature, so the slope at extreme assay temperatures may be steeper than at moderate ones and there is some suggestions that this could be true (Jørgensen *et al.* 2021; Kingsolver & Umbanhowar 2018).

When any shape parameter depends on T the asymmetry correction itself becomes a function of T :

$$\log_{10} \text{LT50}(T) = \beta_0 + \beta_1 (T - \bar{T}) + \frac{1}{k(T)} \log \frac{u(T)-0.5}{0.5-\ell(T)}. \quad (\text{S3})$$

The midpoint piece $\beta_0 + \beta_1 (T - \bar{T})$ is still a straight line in T , but the asymmetry-correction piece now varies with T through $k(T), u(T), \ell(T)$. Their sum is therefore non-linear in T . Two consequences follow, and they affect z and $CT_{max_{1hr}}$ differently: z becomes a *function* of temperature (no longer a single slope but a local value at each T). In contrast, $CT_{max_{1hr}}$ remains a *single* number but its value just shifts because the underlying curve is now bent rather than straight.

- **z becomes a function of temperature.** Since z is defined by the local rate of change of $\log_{10} \text{LT50}$ with T , $z(T) = -1 / \frac{d \log_{10} \text{LT50}(T)}{dT}$. In other words, the local slope of $\log_{10} \text{LT50}$ at any given T varies across the assay range because the shape of the dose-response curve changes with T . This curvature leads to the infinitesimal rates (local slopes) changing which affects the overall sensitivity, z . In these cases, it's better to estimate the local slope first. An overall z can then be obtained by averaging the local slopes if a single summary is needed (pooling the posteriors).
- **CT_{max} has no closed form.** We now need to solve $\log_{10} \text{LT50}(T) = \log_{10} t_{\text{ref}}$ for T per posterior draw — recovered numerically by predicting $\log_{10} \text{LT50}(T)$ from the model on a fine T grid and inverting with `approx()`. The output is the same kind of posterior we already have, just produced numerically rather than algebraically.

11.0.6.1 A numerical recipe for z and CT_{max}

The simplest way to recover z and $CT_{max_{1hr}}$ from a fitted model — whether shape parameters are constant or vary with T — is to **predict survival probabilities from the model** on a fine (T, t) grid and interpolate. We can use `tidybayes::add_epred_draws()` for the prediction and base R's `approx()` for the interpolation. The same code runs whether the fit has constant shape parameters or shape parameters that vary with T .

```
# Temperatures at which we'll evaluate z. When shape parameters depend on T,
# z is itself a function of T, so we compute it at each of the original assay
# temperatures and report the full set rather than a single number.
```

```

T_eval <- temps
half_step <- 0.5 # half-width (°C) of the central finite difference window

# Build a (T, t) grid covering the assay window. The columns log10_t, T_c, and
# n must match the variables the fitted model expects on its input side - they
# are exactly the same as the columns of `design` used to fit the model.
pred_grid <- tidyr::expand_grid(
  T = seq(min(temps) - 1, max(temps) + 1, by = 0.25),
  t = 10 ^ seq(log10(0.5), log10(800), length.out = 200)
) |>
  dplyr::mutate(
    log10_t = log10(t),
    T_c      = T - true_params$T_bar,
    n        = 1
  )

# Now, from the fitted model, draw posterior expected survival probabilities
# at every point in the (T, t) grid. `add_epred_draws()` returns a long tibble
# with one row per (grid point × posterior draw); the column .epred is the
# predicted survival probability under that draw. ndraws keeps it manageable.
epred <- pred_grid |>
  tidybayes::add_epred_draws(fit_bin, ndraws = 500)

# For each (draw, T) we find the duration t at which predicted survival
# crosses 0.5 - that's log10(LT50)(T) for that draw. Linear interpolation
# through the grid via approx() avoids any need for root-finding.
lt50_T <- epred |>
  dplyr::group_by(.draw, T) |>
  dplyr::summarise(
    log10_lt50 = approx(.epred, log10_t, xout = 0.5)$y,
    .groups    = "drop"
  )

# Per posterior draw, compute z at each evaluation temperature in T_eval via
# a central finite difference on the lt50_T grid. For each T_at we approx()
# log10(LT50) at T_at ± half_step and divide the difference by the full step
# (2 * half_step, in °C) to get the slope; z = -1 / slope.
z_per_draw <- purrr::map_dfr(T_eval, function(T_at) {
  lt50_T |>
    dplyr::group_by(.draw) |>
    dplyr::summarise(
      T_at = T_at,
      z     = -1 / ((approx(T, log10_lt50, xout = T_at + half_step)$y -
                    approx(T, log10_lt50, xout = T_at - half_step)$y) /
                    (2 * half_step)),
      .groups = "drop"
    )
})

```

```

})

# Per posterior draw, CTmax_1hr is the temperature at which log10(LT50) =
# log10(60). It is a single number per draw (does not depend on T_eval).
ctmax_per_draw <- lt50_T |>
  dplyr::group_by(.draw) |>
  dplyr::summarise(
    CTmax_1hr = approx(log10_lt50, T, xout = log10(60))$y,
    .groups = "drop"
  )

# Posterior summary of z at each evaluation temperature: median and 95% CrI.
z_per_draw |>
  dplyr::group_by(T_at) |>
  dplyr::summarise(
    z_med = median(z, na.rm = TRUE),
    z_lo = quantile(z, 0.025, na.rm = TRUE),
    z_hi = quantile(z, 0.975, na.rm = TRUE),
    .groups = "drop"
  )

```

```

# A tibble: 5 x 4
  T_at z_med z_lo z_hi
<dbl> <dbl> <dbl> <dbl>
1    30  6.67  6.56  6.79
2    32  6.67  6.56  6.79
3    34  6.67  6.56  6.79
4    36  6.67  6.56  6.79
5    38  6.67  6.56  6.79

```

```

# Posterior summary of CTmax_1hr.
ctmax_per_draw |>
  dplyr::summarise(
    CTmax_1hr_med = median(CTmax_1hr, na.rm = TRUE),
    CTmax_1hr_lo = quantile(CTmax_1hr, 0.025, na.rm = TRUE),
    CTmax_1hr_hi = quantile(CTmax_1hr, 0.975, na.rm = TRUE)
  )

```

```

# A tibble: 1 x 3
  CTmax_1hr_med CTmax_1hr_lo CTmax_1hr_hi
      <dbl>         <dbl>         <dbl>
1      32.1         32.1         32.2

```

The posterior medians of z at each of the five assay temperatures are essentially identical to one another, and the common value matches the closed-form posterior median of z from Section 11.0.4 — exactly as expected for a fit with linear $\text{mid}(T)$ and constant shape parameters, where z does

not depend on temperature. This flatness is the sanity check; if the same recipe is applied to a fit that lets the shape parameters vary with T (e.g. $k \sim T_c$), the same output will show z varying across the assay range, as it should. $CT_{max_{1hr}}$ likewise reproduces the closed-form posterior in `draws_bin_d` from Section 11.0.4.

11.0.6.2 Demonstrating changes with temperature-varying shape parameters

To make the points above concrete, we re-simulate beta-binomial data on the same factorial design as the simple case, but now both the dose-response steepness k and the upper asymptote u vary linearly with assay temperature. We then fit the joint Bayesian 4PL with the corresponding `bf()` formula and apply the same predict-and-interpolate recipe.

```
# (Re-defined here so this chunk can be run independently of the earlier
# simulation. Same values as the original `sim-design` chunk in @sec-sim-step2.)
temps      <- c(30, 32, 34, 36, 38)          # assay temperatures (°C)
durations   <- c(1, 5, 15, 45, 135, 405)     # exposure durations (min)
n_rep       <- 30
n_per_rep   <- 30
phi         <- 5                             # beta-binomial dispersion

design <- tidyr::expand_grid(
  T    = temps,
  t    = durations,
  rep = seq_len(n_rep)
) |>
  dplyr::mutate(
    log10_t = log10(t),
    T_c     = T - 34,                      # mean-centred T (T_bar = 34)
    n       = n_per_rep
  )

# Truth is defined on a constrained raw scale. This keeps ell in [0, 0.49],
# u in [0.51, 1], and k positive at every assay temperature. Those constraints
# avoid the label-switching and invalid-curve geometry that make identity-scale
# 4PL fits mix poorly when u and k vary with temperature.
true_params_ext <- list(
  ell_raw = qlogis(0.05 / 0.49),          # ell = 0.05
  u_raw_0 = qlogis((0.92 - 0.51) / 0.49), # u = 0.92 at T_bar
  u_raw_T = -0.20,                        # u declines as T rises
  log_k_0  = log(8),                      # k = 8 at T_bar
  log_k_T  = 0.10,                        # k rises as T rises
  m_beta0  = 1.5,
  m_beta1  = -0.15,
  T_bar    = 34
)

# Build per-cell true survival probabilities and draw beta-binomial counts.
```

```

# (mutate keeps all columns of `design`; mid and p_true from the original
# sim are overwritten in place with the new T-varying values.)
sim_bb_ext <- design |>
  dplyr::mutate(
    ell    = 0.49 * plogis(true_params_ext$ell_raw),
    u      = 0.51 + 0.49 * plogis(true_params_ext$u_raw_0 +
                                   true_params_ext$u_raw_T * T_c),
    k      = exp(true_params_ext$log_k_0 + true_params_ext$log_k_T * T_c),
    mid    = true_params_ext$m_beta0 + true_params_ext$m_beta1 * T_c,
    p_true = ell +
              (u - ell) /
              (1 + exp(k * (log10_t - mid))),
    p_draw = rbeta(dplyr::n(), p_true * phi, (1 - p_true) * phi),
    y      = rbinom(dplyr::n(), size = n, prob = p_draw)
  )

```

Lets have a look at the data along with truth (Figure S5):

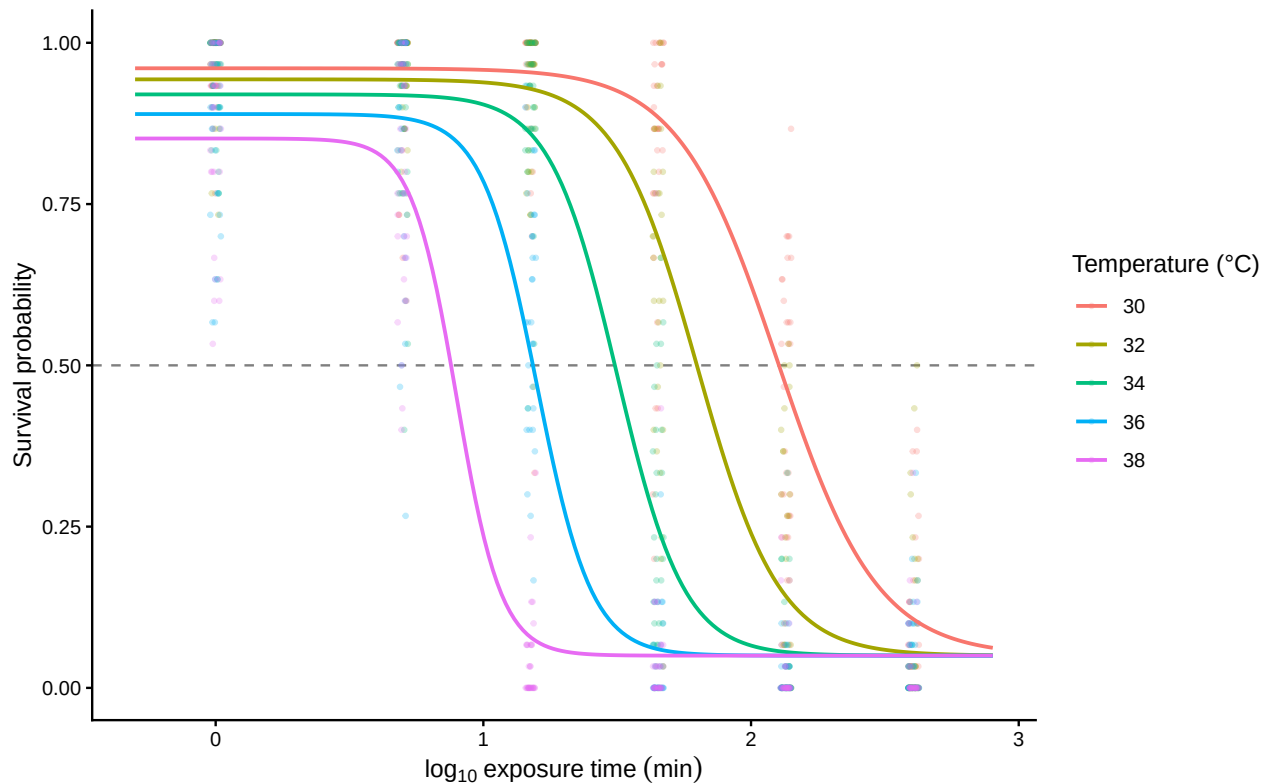


Figure S5: True 4PL dose-response surface (solid lines) for the extended simulation, where k rises and u falls linearly with assay temperature, overlaid on the simulated beta-binomial counts (points). One coloured curve and a corresponding cloud of points per assay temperature; the horizontal dashed line marks 50% survival.

Now that we have the simulated data, let's fit the model. The important change from the simple-model fit is that the four 4PL parameters are no longer fit directly on the identity scale — each

one needs a different transformation because each has a different constraint:

Parameter	Natural-scale constraint	Transform used
<code>ell</code> (lower asymptote)	$[0, 0.49]$	<code>0.49 * inv_logit(ellraw)</code> — squashes raw real to $(0, 0.49)$
<code>u</code> (upper asymptote)	$[0.51, 1]$	<code>0.51 + 0.49 * inv_logit(uraw)</code> — squashes raw real to $(0.51, 1)$
<code>k</code> (slope)	> 0	<code>exp(logk)</code> — squashes raw real to $(0, \infty)$
<code>mid</code> (midpoint)	unconstrained	identity (no transform)

The asymptotes use the **disjoint** intervals $[0, 0.49]$ and $[0.51, 1]$ (rather than each covering all of $[0, 1]$) to prevent MCMC sampling issues where upper and lower asymptotes flip during MCMC sampling which can cause convergence issues — to help here we constrain them and estimate on the logit scale. The 0.02 gap at $p = 0.5$ guarantees $u > \ell$ always, so the chains can't oscillate — which matters here because u and ℓ can vary across temperatures. The cost is hard-bounding the asymptotes a few percentage points away from 0.5 — fine for survival data where genuine LT_{50} assays don't sit at the boundary anyway. We also constrain k to be positive, but on a log scale via `exp(logk)`. Because u , ℓ and k have different transformations and meanings, we apply each transformation per parameter inside the formula before they are combined — as opposed to a single link function on the linear predictor as is typical in GLMMs.

We just need to remember in our fits that these values are transformed so we need to back-transform to make sense of the coefficients.

```
# Disable Quarto chunk caching here so brms's own `file = ...` /
# `file_refit = "on_change"` mechanism is the sole source of truth for whether
# to refit. Without this, the project default `cache: true` can hold an old
# fit object even after the formula or priors change.

# Same 4PL likelihood as before, but with constrained transformations:
#   ell = 0.49 * inv_logit(ellraw)
#   u   = 0.51 + 0.49 * inv_logit(uraw)
#   k   = exp(logk)
# This guarantees ell < 0.5, u > 0.5, u > ell, and k > 0.
bform_ext <- bf(
  y | trials(n) ~
    0.49 * inv_logit(ellraw) +
    ((0.51 + 0.49 * inv_logit(uraw)) -
     0.49 * inv_logit(ellraw)) /
    (1 + exp(exp(logk) * (log10_t - mid))),
  ellraw ~ 1,
  uraw ~ T_c,
  logk ~ T_c,
  mid ~ T_c,
  nl = TRUE,
```

```

    family = beta_binomial(link = "identity")
  )

priors_ext <- c(
  brms::prior_string(
    sprintf("normal(%f, 0.50)", true_params_ext$ell_raw),
    nlpar = "ellraw", coef = "Intercept"
  ),
  brms::prior_string(
    sprintf("normal(%f, 0.50)", true_params_ext$u_raw_0),
    nlpar = "uraw", coef = "Intercept"
  ),
  brms::prior_string(
    sprintf("normal(%f, 0.10)", true_params_ext$u_raw_T),
    nlpar = "uraw", coef = "T_c"
  ),
  brms::prior_string(
    sprintf("normal(%f, 0.35)", true_params_ext$log_k_0),
    nlpar = "logk", coef = "Intercept"
  ),
  brms::prior_string(
    sprintf("normal(%f, 0.05)", true_params_ext$log_k_T),
    nlpar = "logk", coef = "T_c"
  ),
  prior(normal(1.5, 0.40), nlpar = "mid", coef = "Intercept"),
  prior(normal(-0.15, 0.04), nlpar = "mid", coef = "T_c")
)

fit_bb_ext <- brm(
  bform_ext, data = sim_bb_ext, prior = priors_ext,
  chains = 4, iter = 2000, cores = 4, seed = 123,
  control = list(adapt_delta = 0.95, max_treedepth = 12),
  file = file.path(models_dir, "sim_4pl_ext_betabinomial_reparam"),
  file_refit = "on_change"
)

```

The printed model summary is now mainly a convergence check because the coefficients live on raw transformed scales. We still want Rhat near 1 and reasonable effective sample sizes, but recovery is easier to inspect after transforming the posterior draws back to natural 4PL parameters.

```

# Force this chunk to re-run on every render so the printed summary always
# reflects the current `fit_bb_ext` object (the project default `cache: true`
# would otherwise hold onto a stale snapshot when the model is refit upstream).
fit_bb_ext

```

Family: beta_binomial


```

Links: mu = identity
Formula: y | trials(n) ~ 0.49 * inv_logit(ellraw) + ((0.51 + 0.49 * inv_logit(uraw)) - 0.49 * :
ellraw ~ 1
uraw ~ T_c
logk ~ T_c
mid ~ T_c
Data: sim_bb_ext (Number of observations: 900)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
total post-warmup draws = 4000

```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
ellraw_Intercept	-2.31	0.11	-2.54	-2.09	1.00	3056	3207
uraw_Intercept	1.56	0.10	1.37	1.75	1.00	2753	2571
uraw_T_c	-0.22	0.03	-0.28	-0.15	1.00	3307	2825
logk_Intercept	2.06	0.07	1.93	2.21	1.00	2700	2474
logk_T_c	0.15	0.02	0.11	0.19	1.00	3038	2790
mid_Intercept	1.50	0.01	1.48	1.53	1.00	3973	3031
mid_T_c	-0.14	0.01	-0.15	-0.13	1.00	4236	3099

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
phi	4.75	0.36	4.08	5.49	1.00	3381	2977

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Now we can check the changes in z and $CT_{max_{1hr}}$ from having temperature impact on multiple parameters of the 4PL using our workflow in the previous section.

```

# Predict-and-interpolate recipe applied to the extended fit. The code is
# identical to @sec-joint-extend-recipe - only the fit changes.
pred_grid_ext <- tidyrr::expand_grid(
  T = seq(min(temps) - 1, max(temps) + 1, by = 0.25),
  t = 10 ^ seq(log10(0.5), log10(800), length.out = 200)
) |>
dplyr::mutate(
  log10_t = log10(t),
  T_c      = T - true_params_ext$T_bar,
  n        = 1
)

epred_ext <- pred_grid_ext |>
tidybayes::add_epred_draws(fit_bb_ext, ndraws = 500)

lt50_T_ext <- epred_ext |>

```

Table S7: Natural-scale parameter recovery from the constrained temperature-varying 4PL. The posterior is transformed from the raw `ellraw`, `uraw`, and `logk` coefficients back to the biologically interpretable asymptotes, steepness, and midpoint at each assay temperature; ϕ is the beta-binomial precision parameter used to generate the overdispersed counts.

Parameter	Temperature (°C)	Truth	Posterior median	Lower	Upper
ell	30	0.05	0.0443	0.0359	0.0539
k	30	5.363	4.3952	3.7429	5.1646
mid	30	2.1	2.0751	2.0288	2.1233
u	30	0.961	0.9603	0.9466	0.9716
ell	32	0.05	0.0443	0.0359	0.0539
k	32	6.55	5.8805	5.2081	6.673
mid	32	1.8	1.7893	1.7585	1.8221
u	32	0.943	0.9412	0.9282	0.953
ell	34	0.05	0.0443	0.0359	0.0539
k	34	8	7.8697	6.8926	9.1089
mid	34	1.5	1.5043	1.4804	1.5293
u	34	0.92	0.9148	0.901	0.9272
ell	36	0.05	0.0443	0.0359	0.0539
k	36	9.771	10.5314	8.8045	12.9561
mid	36	1.2	1.2187	1.1896	1.2488
u	36	0.89	0.8793	0.859	0.898
ell	38	0.05	0.0443	0.0359	0.0539
k	38	11.935	14.0821	10.9943	18.8182
mid	38	0.9	0.9331	0.8903	0.978
u	38	0.852	0.8355	0.8005	0.8673
phi	all	5	4.7371	4.0814	5.4931

```

dplyr::group_by(.draw, T) |>
dplyr::summarise(
  log10_lt50 = approx(.epred, log10_t, xout = 0.5)$y,
  .groups    = "drop"
)

z_per_draw_ext <- purrr::map_dfr(T_eval, function(T_at) {
  lt50_T_ext |>
  dplyr::group_by(.draw) |>
  dplyr::summarise(
    T_at = T_at,
    z     = -1 / ((approx(T, log10_lt50, xout = T_at + half_step)$y -
                  approx(T, log10_lt50, xout = T_at - half_step)$y) /
                (2 * half_step)),
    .groups = "drop"
  )
})

ctmax_per_draw_ext <- lt50_T_ext |>
dplyr::group_by(.draw) |>
dplyr::summarise(
  CTmax_1hr = approx(log10_lt50, T, xout = log10(60))$y,
  .groups    = "drop"
)

# Posterior summaries: z at each assay temperature, and CTmax_1hr.
z_summary_ext <- z_per_draw_ext |>
dplyr::group_by(T_at) |>
dplyr::summarise(
  z_med = median(z, na.rm = TRUE),
  z_lo  = quantile(z, 0.025, na.rm = TRUE),
  z_hi  = quantile(z, 0.975, na.rm = TRUE),
  .groups = "drop"
)

ctmax_summary_ext <- ctmax_per_draw_ext |>
dplyr::summarise(
  CTmax_1hr_med = median(CTmax_1hr, na.rm = TRUE),
  CTmax_1hr_lo  = quantile(CTmax_1hr, 0.025, na.rm = TRUE),
  CTmax_1hr_hi  = quantile(CTmax_1hr, 0.975, na.rm = TRUE)
)

z_summary_ext

```

```

# A tibble: 5 x 4
  T_at z_med z_lo z_hi
<dbl> <dbl> <dbl> <dbl>

```

1	30	6.79	6.41	7.31
2	32	6.82	6.44	7.36
3	34	6.85	6.47	7.41
4	36	6.88	6.48	7.44
5	38	6.90	6.49	7.47

```
ctmax_summary_ext
```

```
# A tibble: 1 x 3
  CTmax_1hr_med CTmax_1hr_lo CTmax_1hr_hi
      <dbl>      <dbl>      <dbl>
1      32.0      31.8      32.2
```

In the simple-model fit (Section 11.0.4), z was essentially the same value at every assay temperature. In this extended fit, the posterior medians of z vary across temperature (albeit slightly): a direct consequence of the temperature-dependent k and u , which bend the $\log_{10} \text{LT50}(T)$ curve.

In contrast, $CT_{max_{1hr}}$ remains a single number per posterior draw, and in this simulation it actually lands very close to the simple-model value ($\sim 32^\circ\text{C}$): the asymmetry correction is near zero at the temperature where the LT50 curve crosses 1 hour, so $CT_{max_{1hr}}$ is locally almost insensitive to the T-varying shape parameters. With more pronounced T-dependence in k and u , or with shape parameters further from the symmetric ($\ell + u \approx 1$) regime, $CT_{max_{1hr}}$ would shift more visibly.

If we now want to get an adjusted z that averages across these temperatures we can do that as follows:

```
# For each posterior draw, average the five local z(T_eval) values into
# a single adjusted z; then summarise across draws.
adjusted_z_ext <- z_per_draw_ext |>
  dplyr::group_by(.draw) |>
  dplyr::summarise(z_mean = mean(z, na.rm = TRUE), .groups = "drop")

adjusted_z_ext |>
  dplyr::summarise(
    z_med = median(z_mean, na.rm = TRUE),
    z_lo = quantile(z_mean, 0.025, na.rm = TRUE),
    z_hi = quantile(z_mean, 0.975, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 3
  z_med z_lo z_hi
  <dbl> <dbl> <dbl>
1  6.85  6.46  7.40
```

Compared to the simple-model $z = 6.67^\circ\text{C}$ (where shape parameters did not vary with temperature), this adjusted z is 6.85°C with 95% credible interval $[6.46, 7.4]^\circ\text{C}$ — slightly lower in point estimate, with the simple-model value sitting near the upper end of the credible interval. With stronger T-dependence in k and u , the bent $\log_{10} \text{LT50}(T)$ curve would have more variation across temperatures and the two summaries would diverge more.

11.0.7 Heat injury accumulation and survival under a realistic field trace

A particularly powerful aspect of TLS models is their ability to translate dynamic temperature time-series from nature and predict how such temperatures lead to heat injury (HI) accumulation towards a threshold (e.g., LT_{50}) (Arnold *et al.* 2025; Jørgensen *et al.* 2021; Noble *et al.* 2026; Ørsted *et al.* 2022; Rezende *et al.* 2020). The joint Bayesian 4PL fit gives us not just point estimates of z and $CT_{max_{1hr}}$ but their full joint posterior, which we can push through the HI integral to obtain a posterior distribution over the predicted HI trajectory. We can also go a step further and use the *full* 4PL surface to predict the cumulative survival fraction $S_{pred}(t)$ across the trajectory — the second-step prediction discussed in the manuscript (Equation 9 of the manuscript).

Below we apply both calculations to a realistic 5-day field temperature trace using the posterior of `fit_bb` derived in Section 11.0.4.

11.0.7.1 A simulated temperature profile across 5 days

To mimic conditions a moderately heat-tolerant ectotherm might experience over a temperate-zone late-spring period, we generate 5 days of hourly temperatures with a sinusoidal diurnal cycle, ranging from 12°C overnight to a midday peak of 24°C. The critical temperature T_c — below which damage is repaired as fast as it accrues and so there is no net damage increase (Jørgensen *et al.* 2021; Ørsted *et al.* 2022) — is set to 20°C. Injury accumulates only during the warmer part of each day. The simulated organism has $CT_{max_{1hr}} \approx 32.1^\circ\text{C}$ and $z \approx 6.7^\circ\text{C}$ (from `true_params` in Section 11.0.1.1), so the daily peak at 24°C remains 8°C below the 1-hour critical limit — moderate, sub-lethal heat stress that builds cumulatively across days.

```
# Hourly temperature trace: 5 days, sinusoidal diurnal cycle
hi_hours <- 0:(24 * 5 - 1)
hi_T_traj <- 18 + 6 * sin(2 * pi * (hi_hours - 6) / 24) # range 12-24 °C
hi_T_c <- 20 # damage threshold (°C)
```

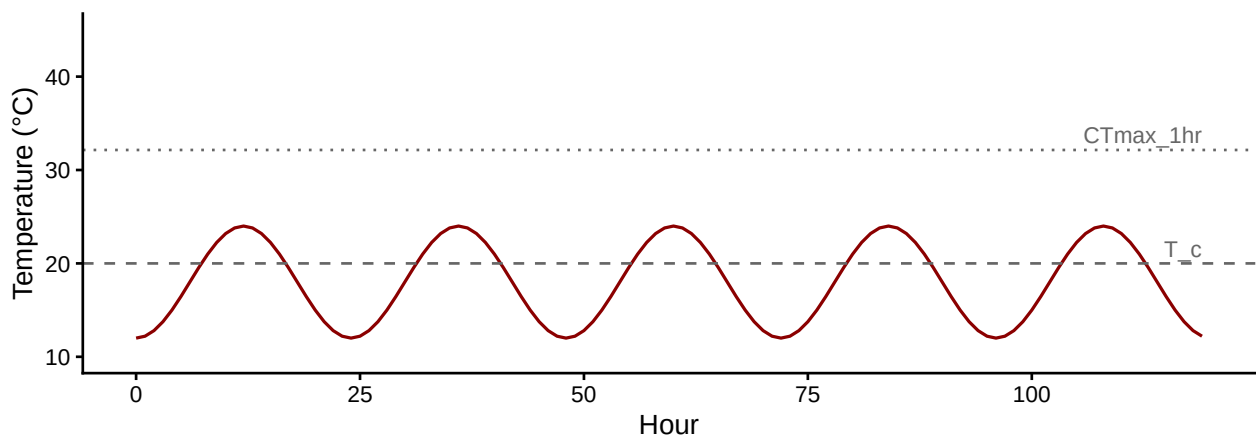


Figure S6: 5-day diurnal temperature trace used for the heat-injury and survival predictions. The dashed line is the damage-accumulation threshold $T_c = 20^\circ\text{C}$; the dotted line is the simulated organism’s $CT_{max_{1hr}} \approx 32.1^\circ\text{C}$ — well above the daily peaks, so accumulation is sub-lethal and gradual.

11.0.7.2 HI accumulation with full posterior uncertainty

For each posterior sample of $(z, CT_{max_{1hr}})$, we evaluate the HI integral (Equation 7 of the manuscript) across the temperature trace. Each sample produces one possible $HI(t)$ trajectory. The posterior median and 95% credible interval at every hour will then summarise the prediction with full parameter uncertainty.

```
# Subsample 500 posterior draws from the beta-binomial fit for speed.
set.seed(20260507)
hi_draws <- posterior::as_draws_df(fit_bb) |>
  dplyr::mutate(
    z      = -1 / b_mid_T_c,
    alpha  = b_mid_Intercept +
              (1 / b_k_Intercept) *
              log((b_u_Intercept - 0.5) / (0.5 - b_ell_Intercept)) -
              b_mid_T_c * true_params$T_bar,
    CTmax_1hr = (log10(60) - alpha) / b_mid_T_c
  ) |>
  dplyr::slice_sample(n = 500)

# Build the (n_hours × n_draws) rate matrix and cumsum down each column.
n_h <- length(hi_T_traj)
n_d <- nrow(hi_draws)

T_mat <- matrix(hi_T_traj, n_h, n_d)
CT_mat <- matrix(hi_draws$CTmax_1hr, n_h, n_d, byrow = TRUE)
z_mat <- matrix(hi_draws$z, n_h, n_d, byrow = TRUE)

# Calculate the rates of HI accumulation from our posterior samples of z and CTmax for the given
rates      <- 100 * 10 ^ ((T_mat - CT_mat) / z_mat)
rates[T_mat <= hi_T_c] <- 0
HI_traj    <- apply(rates, 2, cumsum)
HI_traj[HI_traj > 200] <- 200 # Cap at 200% because survival can't go beyond that.

# Create a summary table that takes the median and 95% CI across the posterior samples for each hour
hi_summary <- tibble::tibble(
  hour      = hi_hours,
  median    = apply(HI_traj, 1, median),
  lower     = apply(HI_traj, 1, quantile, probs = 0.025),
  upper     = apply(HI_traj, 1, quantile, probs = 0.975)
)
```

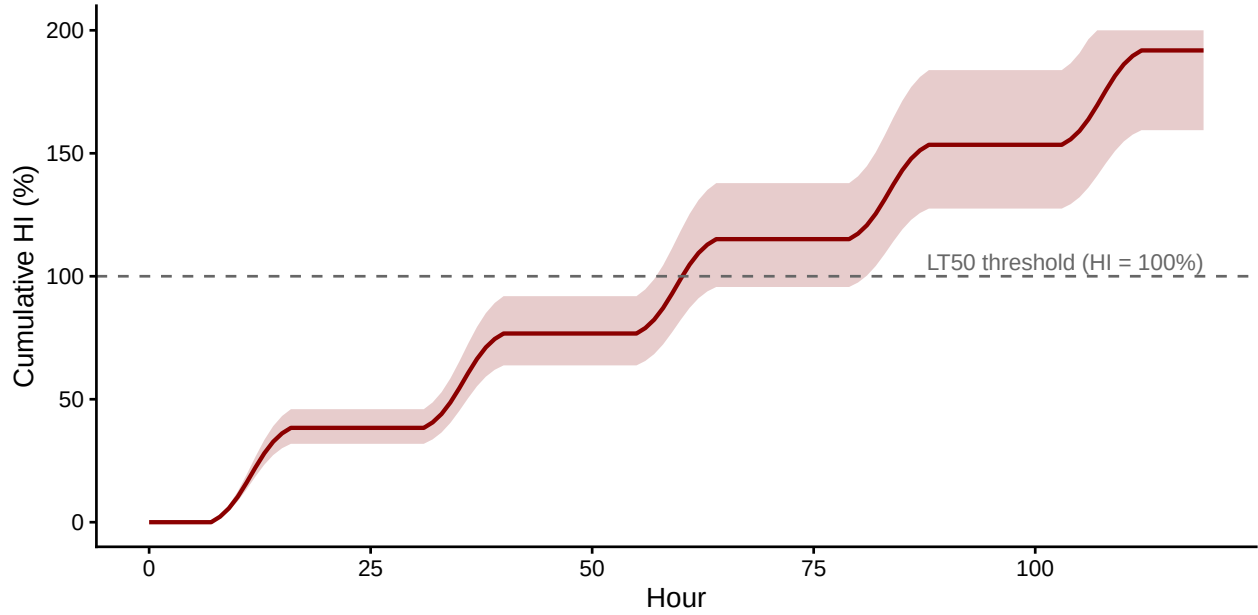


Figure S7: Posterior cumulative heat-injury trajectory under the 5-day field trace. Solid line is the posterior median across 500 draws of $(z, CT_{max_{1hr}})$ from `fit_bb`; shaded band is the 95% credible interval. The dashed line marks the LT_{50} threshold ($HI = 100\%$), which the population is predicted to cross by day 3.

By the end of day 5 the posterior median HI is 192% (95% CI: 159% to 200%) — i.e., the population has accumulated roughly $1.9 \times$ the lethal dose under this exposure.

11.0.7.3 Predicting survival from the full 4PL surface

The HI integral compresses the trajectory into a single threshold-crossing summary. The joint Bayesian 4PL fit also lets us predict the *full* survival fraction $S_{\text{pred}}(t)$ at every time point along the trace, by accumulating effective exposure time at a reference temperature and reading survival off the dose-response surface at the cumulative effective time (Equation 9 of the manuscript). We use $T_{\text{ref}} = \bar{T} = 34^\circ\text{C}$ — the grand-mean temperature of the original assays — so the surface is evaluated at the most data-rich part of the dose-response.

```
T_ref <- true_params$T_bar

# For each draw, accumulate effective minutes at T_ref, then evaluate the 4PL.
S_traj <- matrix(0, n_h, n_d)
for (d in seq_len(n_d)) {
  z_d      <- hi_draws$z[d]
  ell_d    <- hi_draws$b_ell_Intercept[d]
  u_d      <- hi_draws$b_u_Intercept[d]
  k_d      <- hi_draws$b_k_Intercept[d]
  beta0_d  <- hi_draws$b_mid_Intercept[d]  # midpoint at T_bar (in log10-min)

  # Effective-time increment in MINUTES per hourly bin at T_ref = T_bar.
```

```

inc_min      <- ifelse(hi_T_traj > hi_T_c,
                      60 * 10 ^ ((hi_T_traj - T_ref) / z_d),
                      0)
tau_eff_min <- cumsum(inc_min)

# 4PL surface evaluated at (T_ref, tau_eff). Note that tau_eff_min can be 0, and log10(0) is :
log10_tau    <- log10(pmax(tau_eff_min, .Machine$double.eps))
S_traj[, d] <- ell_d + (u_d - ell_d) /
              (1 + exp(k_d * (log10_tau - beta0_d)))
}

S_summary <- tibble::tibble(
  hour    = hi_hours,
  median  = apply(S_traj, 1, median),
  lower   = apply(S_traj, 1, quantile, probs = 0.025),
  upper   = apply(S_traj, 1, quantile, probs = 0.975)
)

```

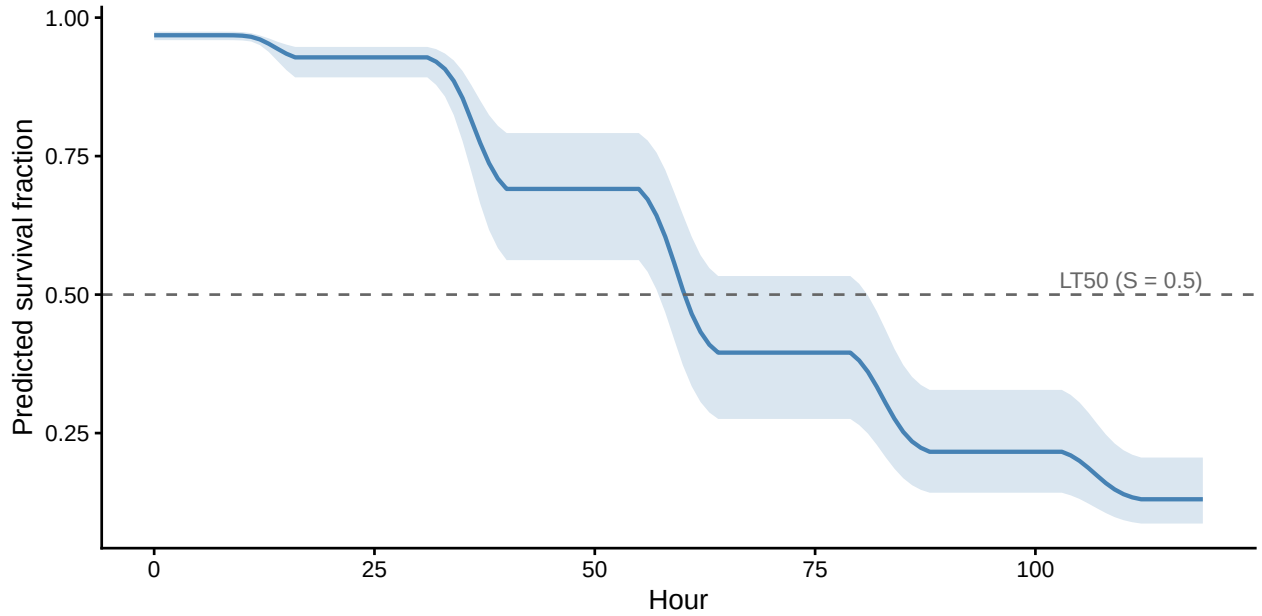


Figure S8: Posterior predicted population-level survival fraction under the 5-day field trace, computed from the model (Equation 9 of the manuscript). Solid line is the posterior median across 500 draws of $(\ell, u, k, \beta_0, \beta_1)$ from `fit_bb`; shaded band is the 95% credible interval. The dashed line marks 50% survival, the threshold the HI integral compresses into a single crossing time.

By the end of day 5 the posterior median predicted survival is 0.13 (95% CI: 0.09 to 0.21). The two predictions tell consistent stories: $S_{\text{pred}}(t)$ falls through 0.5 at the same time that $HI(t)$ rises through 100 %, but $S_{\text{pred}}(t)$ provides the entire cumulative survival curve. Survival can then be used directly for population-dynamical models, which has been suggested by Noble *et al.* (2026).

12 References

- Arnold, P.A., Noble, D.W.A., Nicotra, A.B., Kearney, M.R., Rezende, E.L., Andrew, S.C., *et al.* (2025). [A framework for modelling thermal load sensitivity across life](#). *Global Change Biology*, 31, e70315.
- Bolker, B.M. (2008). *Ecological models and data in R*. Princeton University Press, Princeton, NJ.
- Buckley, L.B., Huey, R.B. & Ma, C.-S. (2025). [How damage, recovery, and repair alter the fitness impacts of thermal stress](#). *Integrative and Comparative Biology*, 65, 1061–1075.
- Burke, D.L., Ensor, J. & Riley, R.D. (2017). [Meta-analysis using individual participant data: One-stage and two-stage approaches, and why they may differ](#). *Statistics in Medicine*, 36, 855–875.
- Bürkner, P.-C. (2017). [brms: An R package for Bayesian multilevel models using Stan](#). *Journal of Statistical Software*, 80, 1–28.
- Carroll, R.J., Ruppert, D., Stefanski, L.A. & Crainiceanu, C.M. (2006). *Measurement error in nonlinear models: A modern perspective*. 2nd edn. Chapman and Hall/CRC.
- Deutsch, C.A., Tewksbury, J.J., Huey, R.B., Sheldon, K.S., Ghalambor, C.K., Haak, D.C., *et al.* (2008). [Impacts of climate warming on terrestrial ectotherms across latitude](#). *Proceedings of the National Academy of Sciences*, 105, 6668–6672.
- Gelman, A. (2006). [Prior distributions for variance parameters in hierarchical models \(comment on article by Browne and Draper\)](#). *Bayesian Analysis*, 1, 515–534.
- Harrison, X.A. (2014). [Using observation-level random effects to model overdispersion in count data in ecology and evolution](#). *PeerJ*, 2, e616.
- Harrison, X.A., Donaldson, L., Correa-Cano, M.E., Evans, J., Fisher, D.N., Goodwin, C.E.D., *et al.* (2018). [A brief introduction to mixed effects modelling and multi-model inference in ecology](#). *PeerJ*, 6, e4794.
- Jørgensen, L.B., Malte, H. & Overgaard, J. (2019). [How to assess Drosophila heat tolerance: Unifying static and dynamic tolerance assays to predict heat distribution limits](#). *Functional Ecology*, 33, 629–642.
- Jørgensen, L.B., Malte, H. & Overgaard, J. (2021). [A unifying model to estimate thermal tolerance limits in ectotherms across static, dynamic and fluctuating exposures to thermal stress](#). *Scientific Reports*, 11, 12840.
- Jørgensen, L.B., Ørsted, M., Malte, H., Wang, T. & Overgaard, J. (2022). [Extreme escalation of heat failure rates in ectotherms with global warming](#). *Nature*, 611, 93–98.
- Kingsolver, J.G. & Umbanhowar, J. (2018). [The analysis and interpretation of critical temperatures](#). *Journal of Experimental Biology*, 221, jeb167858.
- Li, Y.-J., Chen, S.-Y., Jørgensen, L.B., Overgaard, J., Renault, D., Colinet, H., *et al.* (2023). [Interspecific differences in thermal tolerance landscape explain aphid community abundance under climate change](#). *Journal of Thermal Biology*, 114, 103583.
- Nakagawa, S. & Santos, E.S.A. (2012). [Methodological issues and advances in biological meta-analysis](#). *Evolutionary Ecology*, 26, 1253–1274.
- Nielsen, O.K., Ritz, C. & Streibig, J.C. (2004). [Nonlinear mixed-model regression to analyze herbicide dose-response relationships](#). *Weed Technology*, 18, 30–37.
- Noble, D.W.A., Mayfield, M.M., Hoffmann, A.A., Chen, Z.-H., Lade, S.J., Bai, X., *et al.* (2026). [A systems modelling approach to predict biological responses to extreme heat](#). *Trends in Ecology & Evolution*, 41, 451–464.
- Ørsted, M., Jørgensen, L.B. & Overgaard, J. (2022). [Finding the right thermal limit: A framework to reconcile ecological, physiological and methodological aspects of CTmax in ectotherms](#). *Journal of Experimental Biology*, 225, jeb244514.

- Ørsted, M., Willot, Q., Olsen, A.K., Kongsgaard, V. & Overgaard, J. (2024). [Thermal limits of survival and reproduction depend on stress duration: A case study of *Drosophila suzukii*](#). *Ecology Letters*, 27, e14421.
- Pinsky, M.L., Eikeset, A.M., McCauley, D.J., Payne, J.L. & Sunday, J.M. (2019). [Greater vulnerability to warming of marine versus terrestrial ectotherms](#). *Nature*, 569, 108–111.
- Pottier, P., Kearney, M.R., Wu, N.C., Gunderson, A.R., Rej, J.E., Rivera-Villanueva, A.N., *et al.* (2025). [Vulnerability of amphibians to global warming](#). *Nature*, 639, 954–961.
- Rezende, E.L., Bozinovic, F., Szilágyi, A. & Santos, M. (2020). [Predicting temperature mortality and selection in natural *Drosophila* populations](#). *Science*, 369, 1242–1245.
- Rezende, E.L., Castañeda, L.E. & Santos, M. (2014). [Tolerance landscapes in thermal ecology](#). *Functional Ecology*, 28, 799–809.
- Ritz, C. (2010). [Toward a unified approach to dose-response modeling in ecotoxicology](#). *Environmental Toxicology and Chemistry*, 29, 220–229.
- Ritz, C., Baty, F., Streibig, J.C. & Gerhard, D. (2015). [Dose-response analysis using R](#). *PLoS ONE*, 10, e0146021.
- Ritz, C. & Streibig, J.C. (2008). *Nonlinear regression with R*. Springer-Verlag, New York.
- Rudemo, M., Ruppert, D. & Streibig, J.C. (1989). [Random-effect models in nonlinear regression with applications to bioassay](#). *Biometrics. Journal of the International Biometric Society*, 45, 349–362.
- Schielzeth, H. (2010). [Simple means to improve the interpretability of regression coefficients](#). *Methods in Ecology and Evolution*, 1, 103–113.
- Smale, D.A., Wernberg, T., Oliver, E.C.J., Thomsen, M., Harvey, B.P., Straub, S.C., *et al.* (2019). [Marine heatwaves threaten global biodiversity and the provision of ecosystem services](#). *Nature Climate Change*, 9, 306–312.
- Snook, R.R., Bretman, A., Dougherty, L.R. & Fricke, C. (2026). [The consequences of rising temperatures for animal fertility](#). *Nature Reviews Biodiversity*, 2, 225–241.
- Sunday, J.M., Bates, A.E. & Dulvy, N.K. (2011). [Global analysis of thermal tolerance and latitude in ectotherms](#). *Proceedings of the Royal Society B: Biological Sciences*, 278, 1823–1830.
- Sunday, J.M., Bates, A.E. & Dulvy, N.K. (2012). [Thermal tolerance and the global redistribution of animals](#). *Nature Climate Change*, 2, 686–690.
- Truebano, M., Fenner, P., Tills, O., Rundle, S. & Rezende, E. (2018). [Thermal strategies vary with life history stage](#). *Journal of Experimental Biology*, 221, jeb171629.
- Villeneuve, A.R. & White, E.R. (2024). [Predicting organismal response to marine heatwaves using dynamic thermal tolerance landscape models](#). *Journal of Animal Ecology*, 94, 1515–1527.
- Wiens, J.J. (2016). [Climate-related local extinctions are already widespread among plant and animal species](#). *PLOS Biology*, 14, e2001104.

13 Session Information

R version 4.5.3 (2026-03-11)

Platform: aarch64-apple-darwin20

locale: C.UTF-8||C.UTF-8||C.UTF-8||C||C.UTF-8||C.UTF-8

attached base packages: *stats*, *graphics*, *grDevices*, *utils*, *datasets*, *methods* and *base*

other attached packages: *pander*(v.0.6.6), *tidybayes*(v.3.0.7), *tibble*(v.3.3.1), *tinytable*(v.0.16.0), *posterior*(v.1.6.1), *brms*(v.2.23.0), *Rcpp*(v.1.1.1-1.1), *ggplot2*(v.4.0.3), *purrr*(v.1.2.2), *tidyr*(v.1.3.2), *dplyr*(v.1.2.1) and *here*(v.1.0.1)

loaded via a namespace (and not attached): *svUnit*(v.1.0.8), *tidyselect*(v.1.2.1), *farver*(v.2.1.2), *loo*(v.2.9.0), *S7*(v.0.2.2), *fastmap*(v.1.2.0), *TH.data*(v.1.1-3), *tensorA*(v.0.36.2.1), *pacman*(v.0.5.1), *digest*(v.0.6.39), *estimability*(v.1.5.1), *lifecycle*(v.1.0.5), *StanHeaders*(v.2.32.10), *processx*(v.3.9.0), *survival*(v.3.8-6), *magrittr*(v.2.0.5), *compiler*(v.4.5.3), *rlang*(v.1.2.0), *tools*(v.4.5.3), *yaml*(v.2.3.12), *knitr*(v.1.51), *labeling*(v.0.4.3), *bridgesampling*(v.1.2-1), *pkgbuild*(v.1.4.8), *curl*(v.7.1.0), *plyr*(v.1.8.9), *RColorBrewer*(v.1.1-3), *multcomp*(v.1.4-28), *abind*(v.1.4-8), *withr*(v.3.0.2), *grid*(v.4.5.3), *stats4*(v.4.5.3), *colorspace*(v.2.1-1), *xtable*(v.1.8-8), *inline*(v.0.3.21), *emmeans*(v.2.0.3), *scales*(v.1.4.0), *MASS*(v.7.3-65), *tinytex*(v.0.59), *cli*(v.3.6.6), *mvtnorm*(v.1.3-7), *rmarkdown*(v.2.31), *generics*(v.0.1.4), *otel*(v.0.2.0), *RcppParallel*(v.5.1.11-2), *reshape2*(v.1.4.5), *rstan*(v.2.32.7), *stringr*(v.1.6.0), *splines*(v.4.5.3), *bayesplot*(v.1.15.0), *parallel*(v.4.5.3), *matrixStats*(v.1.5.0), *vctrs*(v.0.7.3), *V8*(v.6.0.4), *Matrix*(v.1.7-4), *sandwich*(v.3.1-1), *jsonlite*(v.2.0.0), *callr*(v.3.7.6), *litedown*(v.0.7), *arrayhelpers*(v.1.1-0), *ggdist*(v.3.3.3), *glue*(v.1.8.1), *codetools*(v.0.2-20), *distributional*(v.0.6.0), *stringi*(v.1.8.7), *gtable*(v.0.3.6), *QuickJSR*(v.1.9.0), *pillar*(v.1.11.1), *htmltools*(v.0.5.9), *Brodingnag*(v.1.2-9), *R6*(v.2.6.1), *rprojroot*(v.2.1.1), *evaluate*(v.1.0.5), *lattice*(v.0.22-9), *backports*(v.1.5.1), *rstantools*(v.2.6.0), *coda*(v.0.19-4.1), *gridExtra*(v.2.3), *nlme*(v.3.1-168), *checkmate*(v.2.3.4), *mgcv*(v.1.9-4), *xfun*(v.0.57), *zoo*(v.1.8-14) and *pkgconfig*(v.2.0.3)